Integrated Sensing and Communication in Next-Generation Wireless Networks

Exploiting Dual-Timescale Techniques, Delay-Doppler Alignment Modulation, and Markov Random Field-Based Approaches



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List of Notations

Symbol	Description
M	Number of transmit antennas
B	System bandwidth (Hz)
$T_s = 1/B$	Symbol duration
$L(t)$	Number of multipaths at time t
$\mathbf{h}(t, \tau)$	Time-varying channel impulse response
$\boldsymbol{\eta}(t)$	Path State Information (PSI)
$\bar{T}$	Path-invariant time
T_c	Channel coherence time
ν_l	Doppler frequency of the l-th path
τ_l	Delay of the l-th path
θ_l	Angle of departure (AoD) of the l-th path
α_l	Complex path gain (excluding Doppler)
$\mathbf{f}_l$	Path-based beamforming vector for the l-th path
$s[n]$	Information-bearing symbol
$\mathbf{a}(\theta)$	Array steering vector
$\chi(d_\tau, d_\nu)$	Ambiguity function
Φ_{ISR}	Integrated sidelobe ratio
γ_s	Sensing output SNR
$\mathbf{x}^r, \mathbf{x}^c$	Location-domain sparse radar/communication channel vectors
$\mathbf{s}^r, \mathbf{s}^c, \bar{\mathbf{s}}$	Support vectors for radar, communication, and joint sparsity
α_q, β_{iq}	Spatially non-stationary MRF parameters
$q(\cdot)$	Variational distribution (Turbo-IF-VBI)

1 Introduction

1.1 Motivation and Background

The fifth-generation (5G) mobile communication networks have revolutionized wireless connectivity by enabling enhanced mobile broadband, massive machine-type communications, and ultra-reliable low-latency communications. However, the vision for sixth-generation (6G) networks extends far beyond traditional communication paradigms, aiming to create a seamless bridge between physical and cyber worlds through intelligent connectivity. Integrated Sensing and Communication (ISAC) has emerged as one of the six major usage scenarios for IMT-2030, poised to serve as foundational infrastructure for real-time digital twins, autonomous systems, and immersive extended reality applications.

The fundamental premise of ISAC lies in the recognition that sensing and communication functionalities share numerous similarities in terms of signal processing algorithms, hardware architectures, and channel characteristics. Rather than treating these functions independently—which leads to inefficient resource utilization—ISAC advocates for their synergistic integration, enabling mutual assistance where communication aids sensing and sensing enhances communication.

1.2 Key Challenges in ISAC

Despite its tremendous potential, ISAC faces several fundamental challenges that must be addressed:

Dual Timescale Mismatch: Communication systems typically require estimation of composite Channel State Information (CSI), which remains approximately static during channel coherence time (on the order of milliseconds). In contrast, sensing applications necessitate estimation of individual path parameters—delays, angles, and Doppler frequencies—collectively termed Path State Information (PSI). These parameters vary at significantly different rates, with PSI changing much more slowly than composite CSI. This timescale disparity has been largely overlooked in prior works.

Waveform Design Trade-offs: Conventional orthogonal frequency division multiplexing (OFDM), while dominant in 4G/5G systems, suffers from high peak-to-average power ratio (PAPR), vulnerability to carrier frequency offset, and susceptibility to Doppler-induced inter-carrier interference (ICI). These limitations become particularly pronounced in high-mobility scenarios and mmWave/THz bands.

Channel Estimation Complexity: Accurate estimation of individual multipath parameters—especially Doppler frequencies—remains challenging due to limited signal duration and the superposition of multiple path components. Conventional compressed sensing approaches face off-grid issues and require prior knowledge of the number of paths.

Joint Sparsity Exploitation: In many practical scenarios, radar targets and com-

munication scatterers partially overlap, creating opportunities for joint processing. However, capturing the resulting two-dimensional joint burst sparsity efficiently requires sophisticated probabilistic models.

1.3 Contributions and Organization

This report synthesizes three complementary approaches to address these challenges, drawing from recent advances in the literature:

1. **Dual-Timescale Exploitation for DDAM-based ISAC:** Introduction of path-invariant time as a novel timescale, enabling efficient PSI estimation and channel training overhead reduction.

2. **DAM-based ISAC with Beamforming Optimization:** Comprehensive performance analysis of Delay Alignment Modulation for sensing, including ambiguity function characterization and optimal beamforming design.

3. **Markov Random Field-based Joint Sensing and Estimation:** Probabilistic modeling of two-dimensional joint burst sparsity using spatially non-stationary MRF, with computationally efficient inverse-free variational Bayesian inference.

The remainder of this report is organized as follows. Section II presents a literature survey of related work. Section III provides problem definition and mathematical formulation. Section IV details the methodologies of the three approaches. Section V discusses practical considerations. Section VI presents results and analysis. Section VII offers discussion of insights and trade-offs. Section VIII concludes the report with future research directions.

2 Literature Survey

2.1 Integrated Sensing and Communication Architectures

ISAC systems can be broadly classified into two principal architectures based on the deployment configuration. Monostatic ISAC systems employ the same node for both transmission and reception, requiring full-duplex capability at either the base station (BS) or user equipment (UE). Communication and sensing tasks correspond to two distinct wireless channels in this configuration. Bi-static ISAC systems, in contrast, exploit multipath signals propagating between the communication BS and UE for environment sensing, allowing communication and sensing to share a common wireless channel.

The literature on bi-static ISAC has explored various aspects of joint environment sensing and channel estimation. Liu et al. considered scenarios where radar targets

also serve as communication scatterers, with the BS transmitting downlink pilots for preliminary target detection followed by UE uplink pilots for refinement. Hong et al. proposed geometry-based models for estimating first and last hop scatterers across different bouncing orders. Wen et al. exploited multipath components for simultaneous position estimation and environment mapping.

2.2 Delay-Doppler Domain Signaling

Recent years have witnessed growing interest in delay-Doppler (DD) domain signaling techniques. Orthogonal Time Frequency Space (OTFS) modulation has attracted significant attention due to its attractive performance in high-mobility scenarios. By modulating data symbols in the DD domain, OTFS spreads each symbol over the entire time-frequency resource, achieving full diversity. However, OTFS suffers from high receiver computational complexity and processing latency due to its per-frame based signal processing.

Delay Alignment Modulation (DAM) and its generalization Delay-Doppler Alignment Modulation (DDAM) represent an alternative paradigm. Unlike OTFS, DDAM transforms a time-frequency double-selective fading channel into an equivalent additive white Gaussian noise (AWGN) channel via delay-Doppler compensation and path-based beamforming. This transformation enables inter-symbol interference (ISI)-free communication without requiring sophisticated channel equalization.

2.3 Compressed Sensing for Channel Estimation

The spatial sparsity of mmWave channels, enabled by large-scale antenna arrays and high carrier frequencies, has motivated the application of compressed sensing (CS) techniques for channel estimation. Orthogonal Matching Pursuit (OMP) and its variants have been widely adopted for sparse channel recovery.

However, conventional CS approaches face several limitations: they typically assume on-grid parameters (delays and angles as integer multiples of resolutions), require prior knowledge of sparsity level, and suffer from performance degradation in off-grid scenarios due to power leakage effects. Distributed compressed sensing (DCS) theory has been applied to exploit common sparsity across multiple measurement vectors, but challenges remain in practical implementation.

2.4 Markov Random Field Models for Structured Sparsity

The inherent structure in wireless channels—where non-zero coefficients appear in bursts corresponding to scattering clusters—has motivated the use of Markov Random Field

(MRF) models. MRFs capture spatial dependencies through neighborhood relationships, making them well-suited for modeling two-dimensional burst sparsity.

The Ising model, a specific form of MRF, has been employed to represent binary support vectors with pairwise interactions. Spatially non-stationary MRFs extend this capability by allowing parameters to vary with position, enabling adaptation to different scattering environments. However, learning MRF parameters remains challenging due to the intractable partition function.

2.5 Research Gaps and Contributions

Based on this literature survey, several research gaps are identified:

1. The dual-timescale nature of wireless channels—where PSI varies more slowly than CSI—has not been systematically exploited for ISAC.

2. Channel estimation for DDAM, which requires precise PSI acquisition, remains unsolved.

3. The sensing performance of DAM, particularly in terms of ambiguity functions and sidelobe levels, lacks comprehensive analysis.

4. Joint exploitation of two-dimensional joint burst sparsity in location-domain channels for simultaneous target localization and channel estimation has not been addressed with computationally efficient algorithms.

This report addresses these gaps through the three approaches detailed in subsequent sections.

3 Problem Definition and Mathematical Formulation

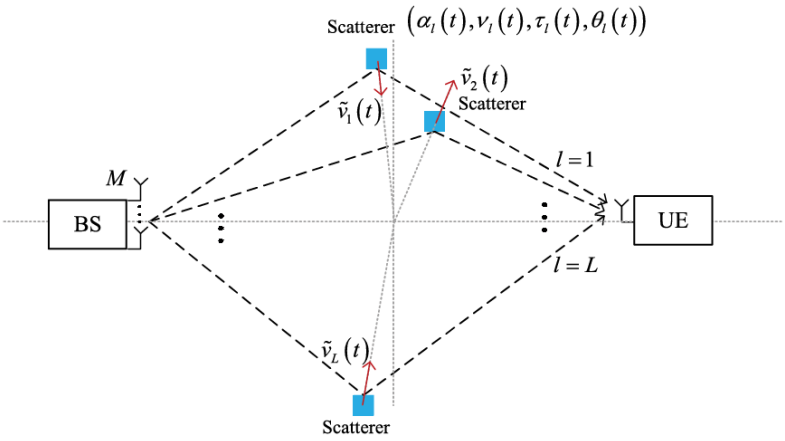


Figure 1: Illustration of the downlink bi-static ISAC system.[1]

3.1 System Model for Bi-static ISAC

Consider a mmWave cellular-based ISAC system where a base station (BS) equipped with $M \gg 1$ transmit antennas and bandwidth B serves a single-antenna UE for downlink communication while concurrently sensing the wireless environment. The baseband equivalent time-varying channel impulse response (CIR) is given by:

$$\mathbf{h}^H(t,\tau) = \sum_{l=1}^{L(t)} \tilde{\alpha}_l(t)\psi(\tau - \tau_l(t))\mathbf{a}^H(\theta_l(t)) \tag{1}$$

where t and τ denote time and delay respectively, $\psi(\cdot)$ is the band-limited pulse shaping response, $L(t)$ is the number of paths (small due to mmWave spatial sparsity), $\tau_l(t)$ is the delay of the lth path, $\mathbf{a}(\theta_l(t)) \in \mathbb{C}^{M\times 1}$ is the transmit array response vector, and the complex channel coefficient is:

$$\tilde{\alpha}_l(t) = \frac{\beta_l e^{-j2\pi f_c \tau_l(t)}}{d_l(t)} \tag{2}$$

with $\beta_l > 0$ the amplitude factor, f_c the carrier frequency, and $d_l(t)$ the propagation distance.

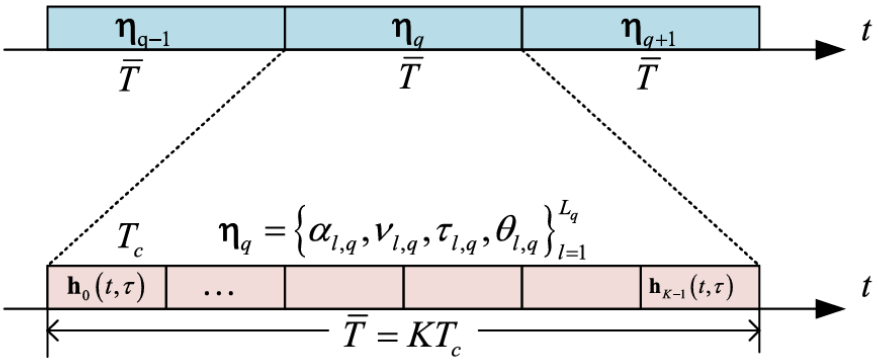


Figure 2: Illustration of the dual timescales for time-varying channels: path-invariant time $\bar{T}$ and channel coherence time T_c. [1]

3.2 Path State Information and Dual Timescales

The Path State Information (PSI) is defined as:

$$\boldsymbol{\eta}(t) = \{\alpha_l(t), \nu_l(t), \tau_l(t), \theta_l(t)\}_{l=1}^{L(t)} \tag{3}$$

where $\nu_l(t)$ is the instantaneous Doppler frequency. Unlike composite CSI, PSI varies much more slowly due to its dependence on geometric channel properties.

Definition (Path-Invariant Time): The path-invariant time $\bar{T}$ is the maximum time interval during which the variation of PSI does not exceed system resolution. During this interval, PSI remains essentially constant:

$$\boldsymbol{\eta}(t) \to \boldsymbol{\eta} = \{\alpha_l, \nu_l, \tau_l, \theta_l\}_{l=1}^{L}, \quad t \in [t_0, t_0 + \bar{T}) \tag{4}$$

Theorem (Path-Invariant Time Bound): For an ISAC system with bandwidth B and $M \gg 1$ transmit antennas, the path-invariant time is:

$$\bar{T} = \min\left\{\frac{c}{3v_{\max}B}, \frac{2R_{\min}}{v_{\max}M}\right\} \approx \min\left\{\frac{c}{3v_{\max}B}, \frac{2R_{\min}}{v_{\max}M}\right\} \tag{5}$$

where $v_{\max}$ is the maximum velocity of scatterers/UE, $R_{\min}$ is the minimum distance between BS and scatterers, and c is the speed of light.

This timescale satisfies $\bar{T} \gg T_c$, where $T_c \approx \xi/\nu_{\max}$ is the channel coherence time.

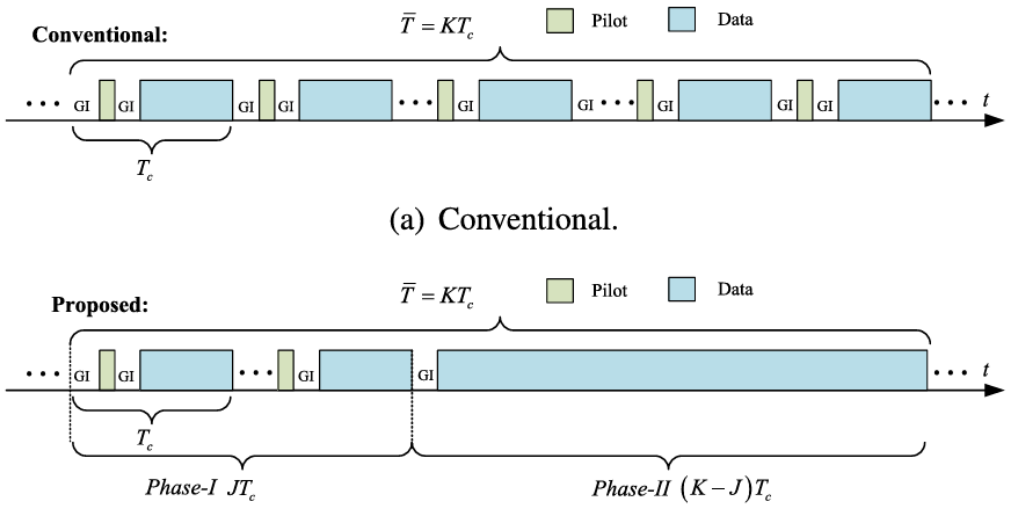


Figure 3: Comparison between conventional pilot-based communication channel estimation and the proposed DDAM-based ISAC frame structure.[1]

3.3 DDAM-based Communication

For DDAM with perfect PSI, the transmitted signal is designed as:

$$\mathbf{x}[n] = \sum_{l=1}^{L} \mathbf{f}_l s[n - \kappa_l] e^{-j2\pi\nu_l n T_s} \tag{6}$$

where $\kappa_l = p_{\max} - p_l$ is the delay pre-compensation, $\mathbf{f}_l$ is the path-based beamforming vector, and $s[n]$ is the information-bearing signal with $\mathbb{E}[|s[n]|^2] = 1$.

The received signal becomes:

$$y[n] = \sum_{l=1}^{L} \sum_{l'=1}^{L} \mathbf{h}_l^H \mathbf{f}_{l'} s[n - \kappa_{l'} - p_l] e^{j2\pi(\nu_l - \nu_{l'})nT_s} + z[n] \tag{7}$$

With zero-forcing beamforming satisfying $\mathbf{h}_l^H \mathbf{f}_{l'} = 0$ for $l \neq l'$, this reduces to an AWGN channel:

$$y[n] = \left(\sum_{l=1}^{L} \mathbf{h}_l^H \mathbf{f}_l \right) s[n - p_{\max}] + z[n] \tag{8}$$

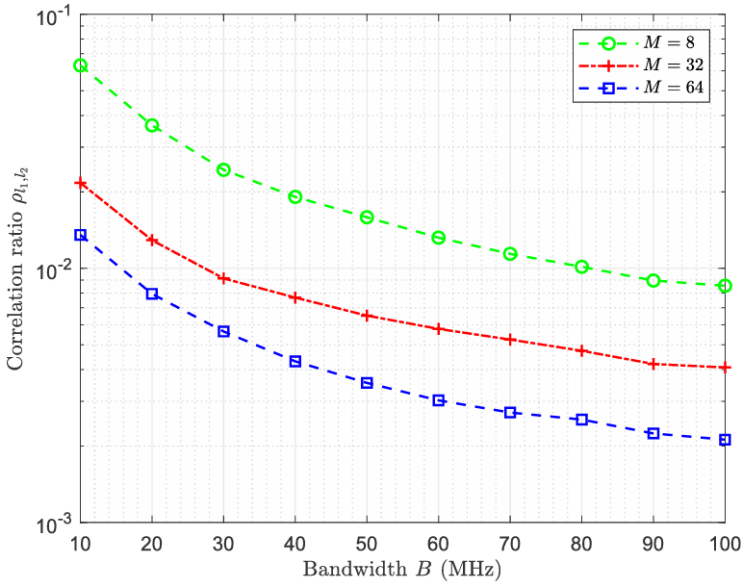


Figure 4: Correlation ratio between two different path components versus bandwidth B for different numbers of transmit antennas M.[1]

3.4 Sensing Performance Metrics

For sensing performance evaluation, we consider:

Ambiguity Function (AF): The normalized AF characterizes resolution and sidelobe properties:

$$\chi(d_\tau, d_\nu) = \frac{\mathbf{a}^H(\theta)\mathbf{F}\mathbf{\Lambda}(d_\tau, d_\nu)\mathbf{F}^H\mathbf{a}(\theta)}{\mathbf{a}^H(\theta)\mathbf{F}\mathbf{\Lambda}(0,0)\mathbf{F}^H\mathbf{a}(\theta)} \tag{9}$$

where $d_\tau = \tau_p - \tau$ and $d_\nu = \nu - \nu_q$ are delay and Doppler differences.

Integrated Sidelobe Ratio (ISR): A metric for distributed scattering susceptibility:

$$\Phi_{\text{ISR}} = \sum_{0<|d_\tau|\leq n_d} |\chi(d_\tau, 0)|^2 \tag{10}$$

Sensing SNR: The output SNR for target detection:

$$\gamma_s = \frac{|\alpha|^2 N \mathbf{a}^H(\theta)\mathbf{F}\mathbf{F}^H\mathbf{a}(\theta)}{\sigma^2} \tag{11}$$

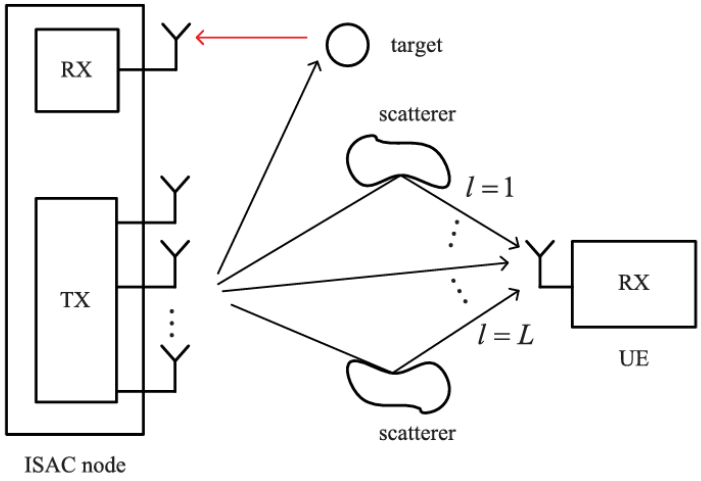


Figure 5: A monostatic ISAC system, where an ISAC node wishes to simultaneously communicate with a UE and sense a target. [2]

3.5 Joint Scattering Environment Sensing Formulation

For the MRF-based approach, the location-domain sparse representation yields:

$$\mathbf{y} = \mathbf{\Phi}(\boldsymbol{\theta})\mathbf{x} + \mathbf{z} \tag{12}$$

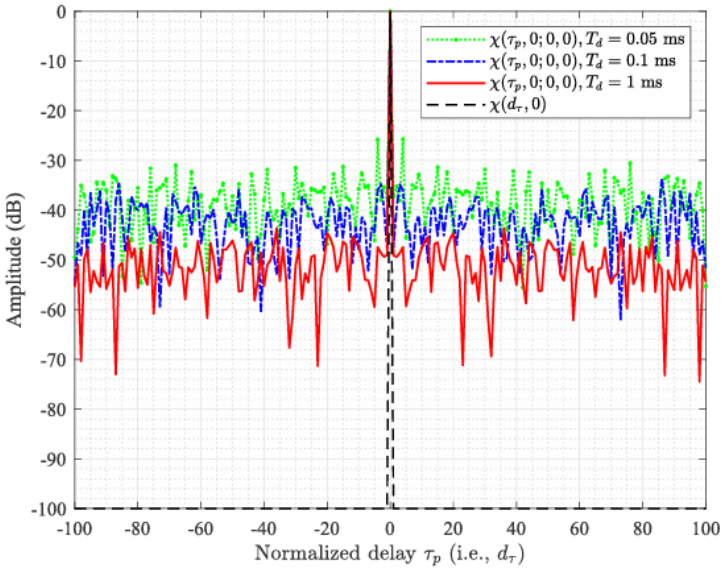
where $\mathbf{x}$ includes radar and communication channel coefficients, and $\mathbf{\Phi}(\boldsymbol{\theta})$ incorporates position-dependent basis functions.

The three-layer sparse prior is:

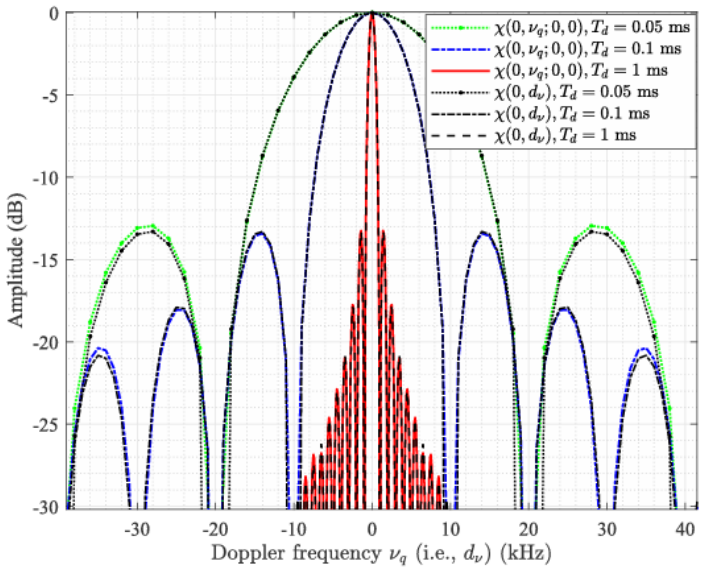
$$p(\mathbf{x}^r, \mathbf{x}^c, \boldsymbol{\rho}^r, \boldsymbol{\rho}^c, \mathbf{s}^r, \mathbf{s}^c, \bar{\mathbf{s}}) = p(\mathbf{s}^r, \mathbf{s}^c, \bar{\mathbf{s}}) p(\boldsymbol{\rho}^r|\mathbf{s}^r) p(\boldsymbol{\rho}^c|\mathbf{s}^c) p(\mathbf{x}^r|\boldsymbol{\rho}^r) p(\mathbf{x}^c|\boldsymbol{\rho}^c) \tag{13}$$

with the joint support modeled by a 2D non-stationary MRF:

$$p(\mathbf{s}) = \frac{1}{Z(\boldsymbol{\zeta})} \exp\left(\sum_{q=1}^{Q}\left(\frac{1}{2}\sum_{i\in\mathcal{N}_q}\beta_{iq}s_i s_q - \alpha_q s_q\right)\right) \tag{14}$$



(a) Doppler-cut AF at $\nu_q = \nu = 0$, i.e., $d_\nu = 0$



(b) Delay-cut AF at $\tau_p = \tau = 0$, i.e., $d_\tau = 0$

Figure 6: Comparison between the normalized AF $\chi(\tau_p, \nu_q; \tau, \nu)$ and the asymptotic AF $\chi(d_\tau, d_\nu)$ of the DAM signal with single-path beamforming. [2]

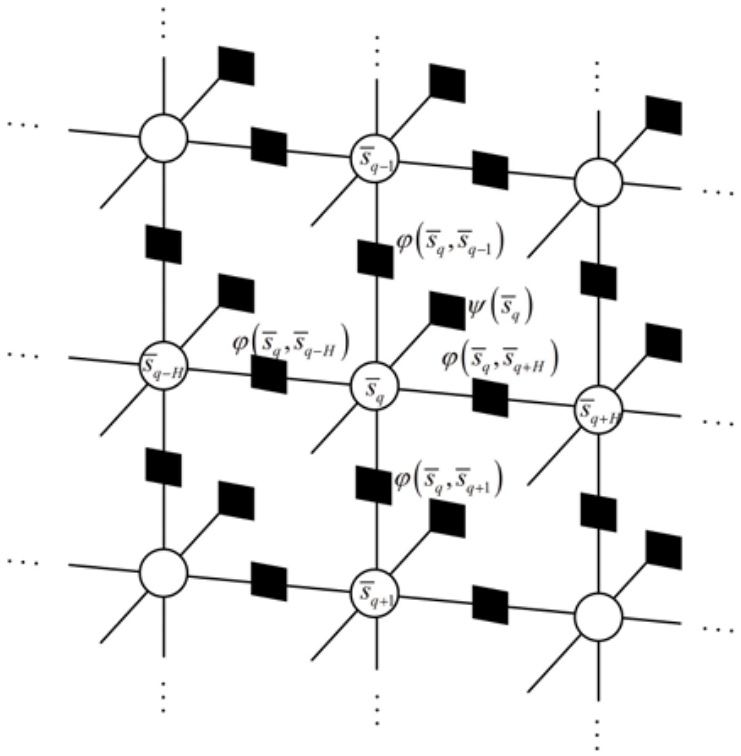


Figure 7: Factor graph of the 4-connected MRF model.[3]

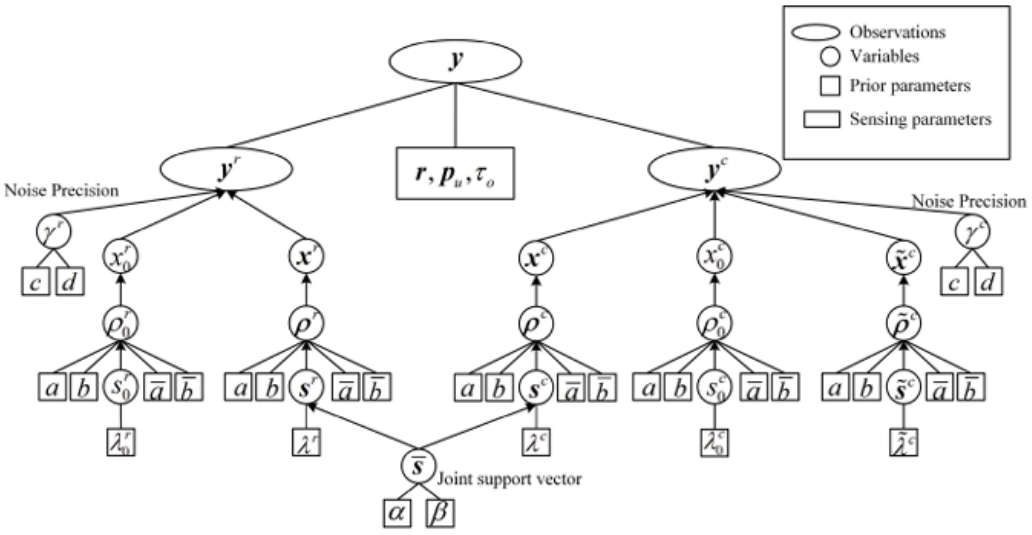


Figure 8: Illustration of the three-layer sparse prior model.[3]

4 Methodology

4.1 Approach 1: Dual-Timescale DDAM-based ISAC

4.1.1 Adaptive Simultaneous OMP with Support Refinement (ASOMP-SR)

The proposed ASOMP-SR algorithm leverages the common sparsity across multiple channel coherence blocks within the path-invariant time. Algorithm 1 presents the complete procedure.

Algorithm 1 ASOMP-SR Algorithm

Require: Received pilots $\mathbf{y}_k$, sensing matrices $\mathbf{\Phi}_k$, threshold ϵ_{th}
Ensure: Estimated channel vectors $\hat{\mathbf{h}}_k$, support set $\hat{\Theta}$

Initialize $J^{(0)} = 0$, $u = 1$
repeat
  $J^{(u)} = J^{(u-1)} + 1$
  Collect $\mathbf{y}_k$ and $\mathbf{\Phi}_k$ for $k = 0, \ldots, J^{(u)} - 1$
  $\mathcal{T}^{(u)} = \emptyset$, $\iota = 1$
  $\mathbf{r}_k^{(0)} = \mathbf{y}_k$ for all k
  repeat
    $\hat{g}_k = \arg\max_g \sum_{k=0}^{J^{(u)}-1} \|[\mathbf{\Phi}_k]_{:,g}^H \mathbf{r}_k^{(\iota-1)}\|$
    $\mathcal{T}^{(u)} = \mathcal{T}^{(u)} \cup \{\hat{g}_k\}$
    $\mathbf{r}_k^{(\iota)} = \mathbf{r}_k^{(\iota-1)} - [\mathbf{\Phi}_k]_{:,\mathcal{T}^{(u)}} [\mathbf{\Phi}_k]_{:,\mathcal{T}^{(u)}}^{\dagger} \mathbf{r}_k^{(\iota-1)}$
    Compute residual difference ϵ_k
    $\iota = \iota + 1$
  until $\epsilon_k \leq \epsilon_{\text{th}}$
  Estimate $\hat{\mathbf{h}}_k^{(u)} = (\mathbf{\Psi}_k^{(u)})^{\dagger} \mathbf{y}_k$
  Perform support refinement (Algorithm 2)
  Compute recovery difference θ_u
  $u = u + 1$
until θ_u does not decrease

The support refinement procedure (Algorithm 2) operates by:

1. Detecting the strongest angular-delay bin for each path
2. Constructing support sets including neighboring bins to account for power leakage
3. Refining estimates by retaining only detected support
4. Terminating when power ratio stops increasing

4.1.2 Doppler Frequency Estimation

After obtaining angular-delay domain estimates $\check{\mathbf{h}}_k$ for $k = 0, \ldots, J-1$, Doppler frequencies are estimated via:

$$\mathbf{u}_l^H = \check{\mathbf{h}}_{l,0}^H \check{\mathbf{H}} \approx \varsigma_l[1, e^{-j2\pi\nu_l T_c}, \ldots, e^{-j2\pi\nu_l (J-1)T_c}] \tag{15}$$

The Doppler estimate is obtained via IDFT:

$$\hat{\omega}_l = \arg\max_{\omega} \|\mathbf{u}_l^H \mathbf{v}(\omega)\|, \quad \hat{\nu}_l = \frac{\hat{\omega}_l}{N_o J T_c} \tag{16}$$

where $\mathbf{v}(\omega) = [1, e^{j2\pi\omega/(N_o J)}, \ldots, e^{j2\pi\omega(J-1)/(N_o J)}]^T$ and N_o controls search grid refinement.

4.2 Approach 2: DAM-based ISAC with Beamforming Optimization

4.2.1 Sensing Performance Analysis

The asymptotic AF for DAM with $N \gg 1$ symbols per CPI exhibits separable structure:

$$\chi(d_\tau, d_\nu) = \begin{cases} 1, & d_\tau = 0, d_\nu = 0 \\ \chi(d_\tau, 0), & d_\tau \neq 0, d_\nu = 0 \\ \chi(0, d_\nu), & d_\tau = 0, d_\nu \neq 0 \\ \chi(d_\tau, 0)\chi(0, d_\nu), & d_\tau \neq 0, d_\nu \neq 0 \end{cases} \tag{17}$$

The Doppler-cut AF is:

$$\chi(0, d_\nu) = e^{j\pi d_\nu (N-1)T_s} \mathrm{asin}(d_\nu, N) \tag{18}$$

where $\mathrm{asin}(d_\nu, N) = \frac{\sin(\pi d_\nu N T_s)}{N \sin(\pi d_\nu T_s)}$ is the aliased sinc function. The peak sidelobe ratio is approximately -13.2 dB.

The delay-cut AF depends on beamforming:

$$\chi(d_\tau, 0) = \frac{\mathbf{a}^H(\theta) \left(\sum_{(i,j)\in\mathcal{S}(d_\tau)} \mathbf{f}_i \mathbf{f}_j^H \right) \mathbf{a}(\theta)}{\mathbf{a}^H(\theta) \sum_{l=1}^{L} \mathbf{f}_l \mathbf{f}_l^H \mathbf{a}(\theta)} \tag{19}$$

4.2.2 Beamforming Optimization Problem

The optimization problem maximizes communication SNR subject to sensing constraints:

$$\text{(P1)}: \quad \max_{\{\mathbf{f}_l\}_{l=1}^L} \left| \sum_{l=1}^{L} \mathbf{h}_l^H \mathbf{f}_l \right|^2 \tag{20}$$

$$\text{s.t.} \quad \mathbf{h}_{l'}^H \mathbf{f}_l = 0, \quad \forall l \neq l' \tag{21}$$

$$\sum_{l=1}^{L} \|\mathbf{f}_l\|^2 \leq P_t \tag{22}$$

$$\frac{|\alpha|^2 N}{\sigma^2} \mathbf{a}^H(\theta) \left(\sum_{l=1}^{L} \mathbf{f}_l \mathbf{f}_l^H \right) \mathbf{a}(\theta) \geq \gamma_{\text{th}} \tag{23}$$

$$\sum_{0<|d_\tau|\leq n_d} \left| \mathbf{a}^H(\theta) \left(\sum_{(i,j)\in\mathcal{S}(d_\tau)} \mathbf{f}_i \mathbf{f}_j^H \right) \mathbf{a}(\theta) \right|^2 \leq \phi_{\text{th}} \left| \mathbf{a}^H(\theta) \sum_{l=1}^{L} \mathbf{f}_l \mathbf{f}_l^H \mathbf{a}(\theta) \right|^2 \tag{24}$$

Using semidefinite relaxation (SDR), the problem is reformulated as:

$$\text{(P2.1)}: \quad \max_{\bar{\mathbf{B}}} \operatorname{Tr}(\bar{\mathbf{H}}\bar{\mathbf{B}}) \tag{25}$$

$$\text{s.t.} \quad \operatorname{Tr}(\bar{\mathbf{Q}}\bar{\mathbf{B}}) \leq P_t \tag{26}$$

$$\operatorname{Tr}(\bar{\mathbf{A}}(\theta)\bar{\mathbf{B}}) \geq \tilde{\gamma}_{\text{th}} \tag{27}$$

$$\sum_{0<|d_\tau|\leq n_d} \left| \operatorname{Tr}(\bar{\mathbf{A}}_Q(\theta, d_\tau)\bar{\mathbf{B}}) \right|^2 \leq \tilde{\phi}_{\text{th}} \tag{28}$$

$$\bar{\mathbf{B}} \succeq 0 \tag{29}$$

This convex optimization can be solved efficiently using CVX, with rank reduction via eigenvalue decomposition and Gaussian randomization.

4.3 Approach 3: MRF-based Joint Sensing and Channel Estimation

4.3.1 Turbo Inverse-Free Variational Bayesian Inference (Turbo-IF-VBI)

The Turbo-IF-VBI algorithm iterates between E-step and M-step:

E-Step: For given parameters $\boldsymbol{\xi}^{(t)}$, approximate marginal posteriors using inverse-free VBI:

The relaxed ELBO is:

$$\tilde{L}(q, \mathbf{w}) = \int_{\mathbf{v}} q(\mathbf{v}) \ln \frac{G(\mathbf{y}, \mathbf{v}, \mathbf{w}; \boldsymbol{\xi})}{q(\mathbf{v})} \tag{30}$$

where G incorporates a quadratic lower bound on the likelihood. The posterior mean and covariance become:

$$\boldsymbol{\mu} = \boldsymbol{\Sigma}\left(\boldsymbol{\Phi}(\boldsymbol{\theta})^H\langle\boldsymbol{\Gamma}\rangle(\mathbf{y} - \boldsymbol{\Phi}(\boldsymbol{\theta})\mathbf{w}) + \langle\boldsymbol{\Gamma}\rangle\mathbf{T}\mathbf{w}\right) \tag{31}$$

$$\boldsymbol{\Sigma} = (\langle\boldsymbol{\Gamma}\rangle\mathbf{T} + \mathrm{diag}(\langle\boldsymbol{\rho}\rangle))^{-1} \tag{32}$$

Note that $\boldsymbol{\Sigma}$ involves only diagonal matrix inversion, avoiding the $\mathcal{O}((1+Q+UV)^3)$ complexity of conventional approaches.

M-Step: Update parameters using gradient ascent with pseudo-likelihood approximation for MRF parameters:

$$\frac{\partial\tilde{Q}_\zeta(\zeta;\boldsymbol{\xi}^{(t)})}{\partial\alpha_q} = \mathbb{E}_{q(\bar{\mathbf{s}}|\mathbf{y};\boldsymbol{\xi}^{(t)})}\left\{-\bar{s}_q + \frac{\exp(2\sum_{i\in\mathcal{N}_q}\beta_{iq}\bar{s}_i) - \exp(2\alpha_q)}{\exp(2\sum_{i\in\mathcal{N}_q}\beta_{iq}\bar{s}_i) + \exp(2\alpha_q)}\right\} \tag{33}$$

$$\frac{\partial\tilde{Q}_\zeta(\zeta;\boldsymbol{\xi}^{(t)})}{\partial\beta_{pq}} = \mathbb{E}_{q(\bar{\mathbf{s}}|\mathbf{y};\boldsymbol{\xi}^{(t)})}\left\{2\bar{s}_p\bar{s}_q - \bar{s}_p\frac{\exp(2\sum_{i\in\mathcal{N}_q}\beta_{iq}\bar{s}_i) - \exp(2\alpha_q)}{\exp(2\sum_{i\in\mathcal{N}_q}\beta_{iq}\bar{s}_i) + \exp(2\alpha_q)}\right\} \tag{34}$$

4.3.2 Message Passing in MRF

The factor graph of the 4-connected MRF model involves variable nodes $\{s_q\}$ arranged in H rows and W columns. Messages are exchanged according to the sum-product rule:

$$\nu_{u_q^r\to s_q} \propto \sum_{s_q^r}\nu_{s_q^r\to u_q^r}\cdot u_q^r(s_q^r, s_q) \tag{35}$$

The approximate marginal posterior is:

$$q(s_q|\mathbf{y};\boldsymbol{\xi}) \propto \nu_{u_q^r\to s_q}\cdot\nu_{u_q^c\to s_q}\cdot\prod_{i\in\mathcal{N}_q}\nu_{i\to s_q}\cdot\psi(s_q) \tag{36}$$

5 Practical Considerations

5.1 Implementation Challenges and Solutions

5.1.1 Computational Complexity

The three approaches present different complexity profiles:

Table 1: Computational Complexity Comparison

Operation	**DDAM-ASOMP-SR**	**DAM-SDR**	**Turbo-IF-VBI**
Per-iteration complexity	$\mathcal{O}(J\bar{L}MP)$	$\mathcal{O}((ML)^3)$	$\mathcal{O}(M\|\mathcal{N}_b\|Q)$
Matrix inversion	None	Required	Avoided via diagonalization
Parallelization potential	High	Medium	High
Memory requirements	Low	High	Medium

The ASOMP-SR algorithm avoids matrix inversion entirely, making it suitable for large-scale antenna arrays. The DAM optimization with SDR requires solving a convex problem but scales with $(ML)^3$, which may be prohibitive for massive MIMO. The Turbo-IF-VBI algorithm eliminates the matrix inverse through the relaxed ELBO, achieving $\mathcal{O}(M|\mathcal{N}_b|Q)$ complexity.

5.1.2 Off-Grid Effects and Power Leakage

In practical systems, true delays and angles rarely lie exactly on the discretized grid points. This off-grid condition causes power leakage across neighboring bins, degrading estimation accuracy. The proposed methods address this through:

1. **Support Refinement (ASOMP-SR):** Retains $V_r \times V_p$ neighboring bins around detected peaks, capturing leaked power.
2. **Dynamic Position Grid (Turbo-IF-VBI):** Updates grid positions $\mathbf{r}$ via gradient ascent, allowing grid points to move toward true positions.
3. **Over-sampled DFT (Doppler estimation):** Uses sampling factor $N_o > 1$ to create finer search grid.

5.1.3 Pilot Overhead and Frame Structure

The dual-timescale property enables significant pilot overhead reduction. Within each path-invariant block of duration $\bar{T} = KT_c$, only $J \ll K$ coherence blocks are used for PSI estimation. The overhead saving is:

$$\text{Overhead Saving} = (K - J)(N_p + 2N_g) - N_g \tag{37}$$

where N_p is pilot length and N_g is guard interval length. For typical parameters ($K = 500$, $J = 10$, $N_p = 100$), this represents approximately 98% reduction in pilot overhead.

5.2 Hardware Implications

5.2.1 Delay Pre-compensation Implementation

Delay pre-compensation in DDAM corresponds to simple sequence shifting in digital baseband processing. For a symbol-spaced system with $T_s = 1/B$, the maximum delay spread determines the required memory. With $B = 100$ MHz and maximum delay of 10μs, approximately 1000 symbol buffers are needed—feasible with modern FPGA/ASIC implementations.

5.2.2 Beamforming Requirements

Path-based beamforming requires $M \times L$ complex multipliers per symbol period. For $M = 64$ and $L = 5$, this is 320 multiplications, comparable to conventional channel-based beamforming. The additional complexity comes from per-path delay compensation, which can be pipelined efficiently.

5.2.3 Power Amplifier Considerations

The lower PAPR of DDAM/DAM compared to OFDM (approximately L vs K for constant modulus beamforming) allows operation closer to amplifier saturation. For $L = 5$ and $K = 2048$, the average transmit power can be increased by a factor of $K/L \approx 410$ (26 dB) before saturation, significantly extending communication and sensing range.

5.3 Environmental Adaptivity

5.3.1 Scattering Environment Dynamics

The spatially non-stationary MRF model adapts to different scattering environments through position-dependent parameters α_q (sparsity) and β_{iq} (cluster size). In dense urban environments with rich scattering, β_{iq} values are larger, creating larger bursts. In rural environments with sparse scattering, α_q values are larger, promoting sparsity.

5.3.2 Mobility Adaptation

The path-invariant time $\bar{T}$ decreases with increasing velocity. The system can adapt by:

1. Monitoring recovery difference θ_u to detect PSI changes
2. Reducing J (number of estimation blocks) when mobility increases
3. Triggering re-estimation when θ_u exceeds threshold

6 Results and Analysis

6.1 Simulation Setup

Common simulation parameters across approaches:

Table 2: Simulation Parameters

Parameter	**Value**
Carrier frequency (f_c)	28-60 GHz
Bandwidth (B)	100 MHz
Number of BS antennas (M)	64
Number of paths (L)	5
UE distance	100 m
Scatterer distances	10-100 m
Maximum Doppler ($\nu_{\max}$)	4-10 kHz
Coherence time (T_c)	0.1 ms
Path-invariant time ($\bar{T}$)	20-50 ms
Noise power	-89 to -94 dBm

6.2 Channel Estimation Performance

6.2.1 NMSE Comparison

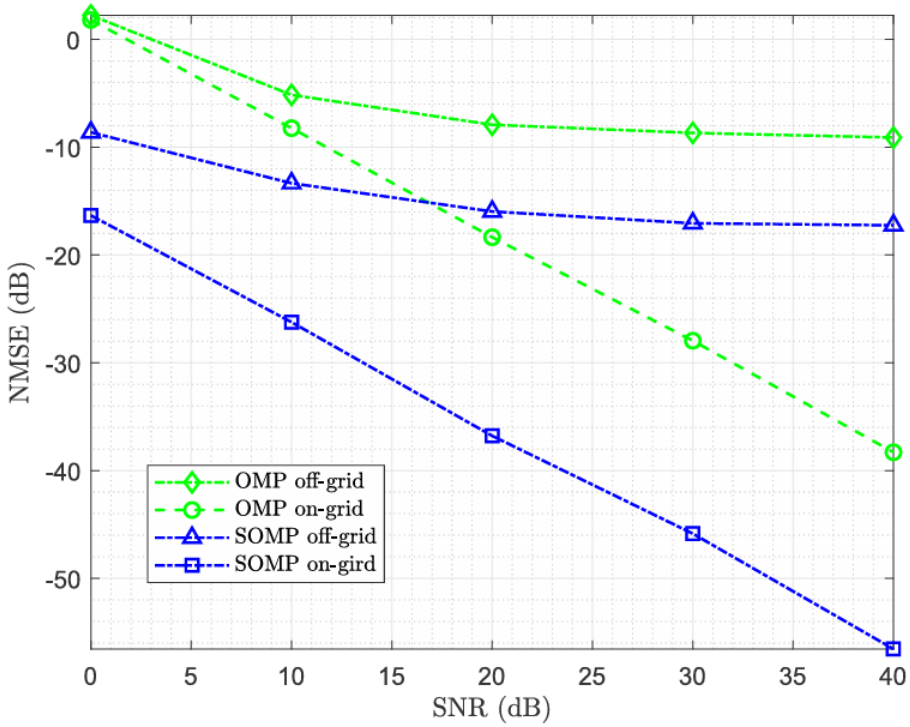


Figure 9: NMSE performance of OMP and SOMP methods versus SNR for both on-grid and off-grid cases.[1]

Key observations:

1. All schemes show decreasing NMSE with increasing SNR, as expected.
2. On-grid performance is significantly better than off-grid due to higher sparsity.
3. SOMP outperforms OMP for both on-grid and off-grid cases, demonstrating the benefit of joint sparsity exploitation across multiple coherence blocks.

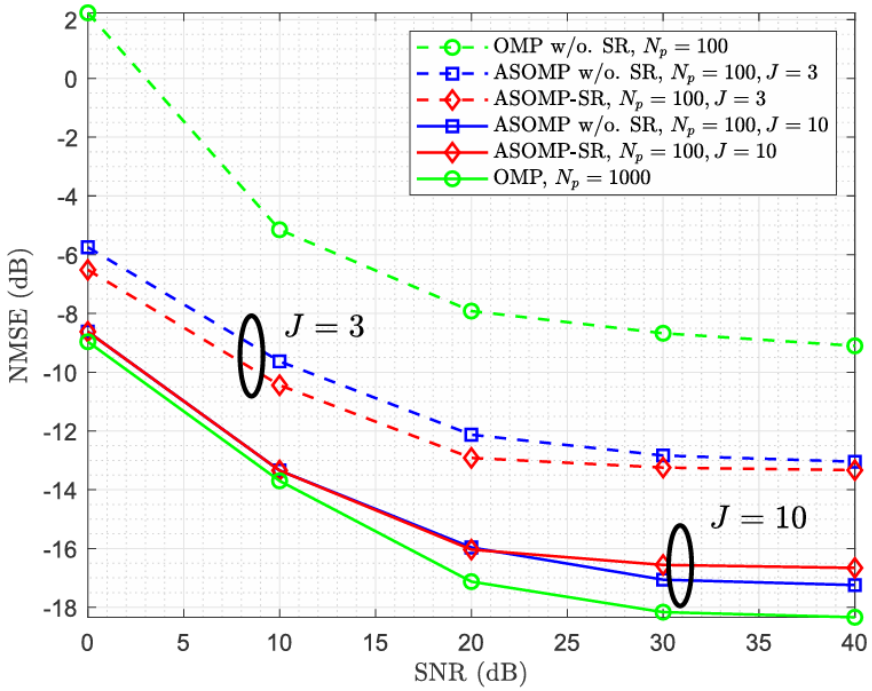
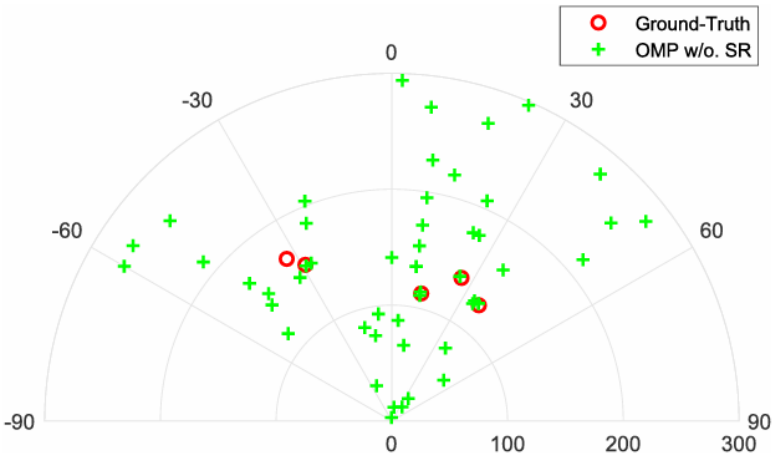


Figure 10: Comparison of OMP and proposed ASOMP-SR methods in terms of NMSE versus SNR for the off-grid case.[1]

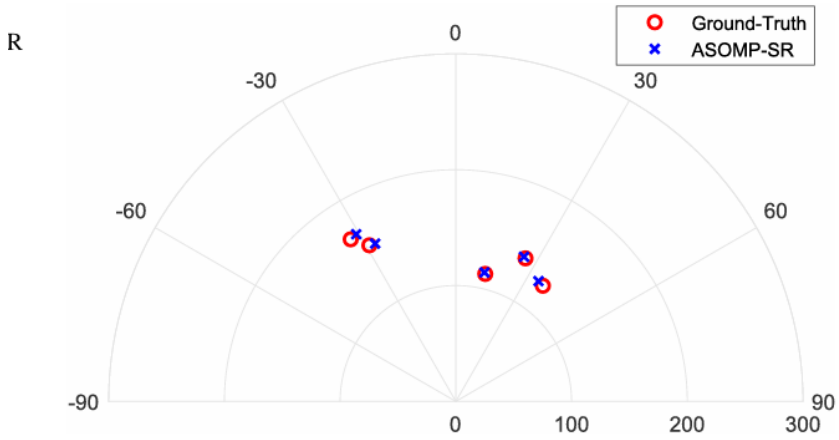
The proposed ASOMP-SR achieves:

- With $J = 10$ and $N_p = 100$, ASOMP matches OMP with $N_p = 1000$—a 10x pilot reduction.
- Support refinement improves NMSE by 3-5 dB across all SNR ranges.
- Performance gain saturates for $J > 10$, indicating diminishing returns from additional blocks.

6.2.2 Path Resolution Capability



(a) Environment sensing of OMP w/o. SR.



(b) Environment sensing of the proposed ASOMP-SR.

Figure 11: Comparison of conventional OMP without SR and the proposed ASOMP-SR in terms of environment sensing performance in off-grid case. [1]

Conventional OMP without SR fails to resolve paths correctly due to power leakage and unknown sparsity level. ASOMP-SR accurately estimates number of paths L, delays τ_l, and AoDs θ_l simultaneously. The SR algorithm correctly identifies $V_r = V_p = 8$ neighboring bins per path, capturing 99% of leaked power.

6.2.3 Doppler Estimation Accuracy

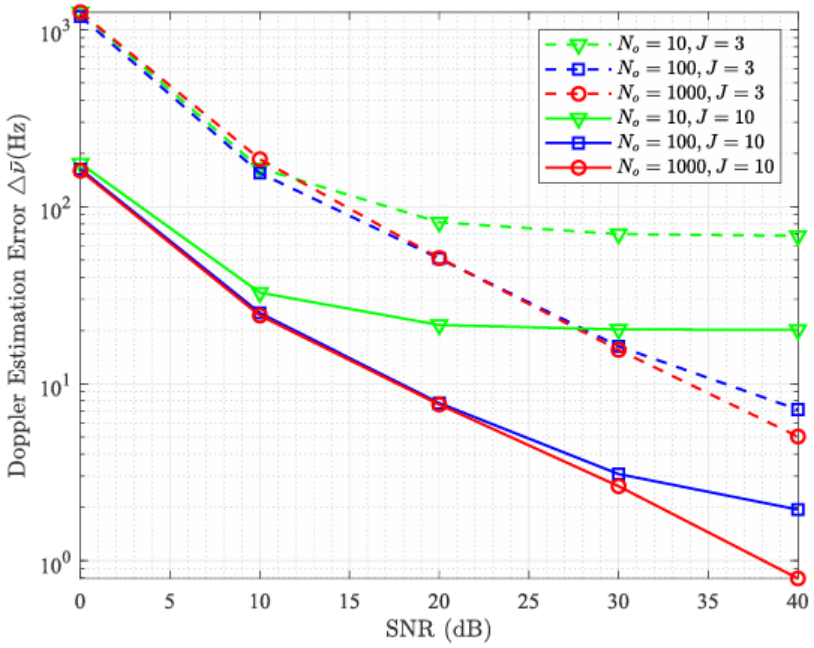
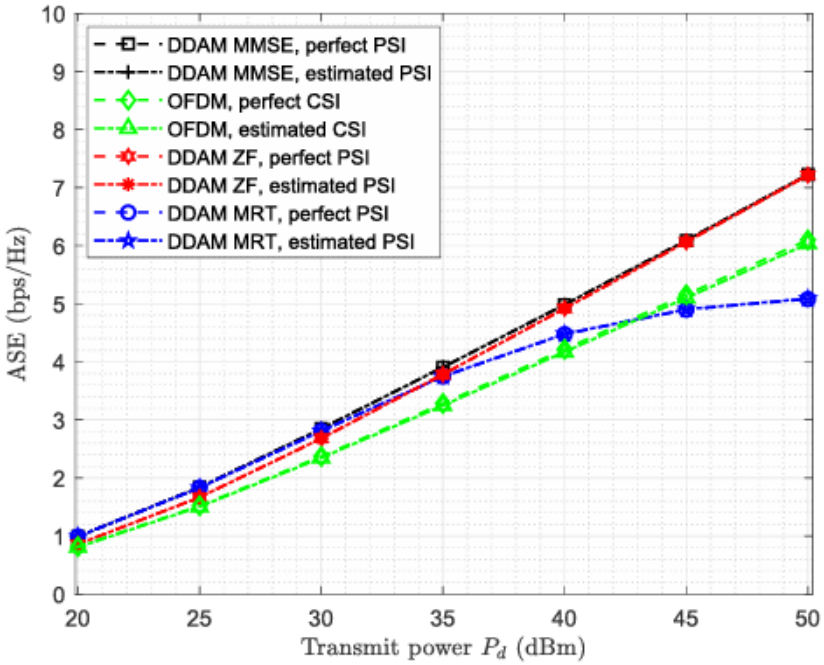


Figure 12: Doppler estimation error versus SNR with different sampling factor $N_o = 10, 100, 1000$.[1]

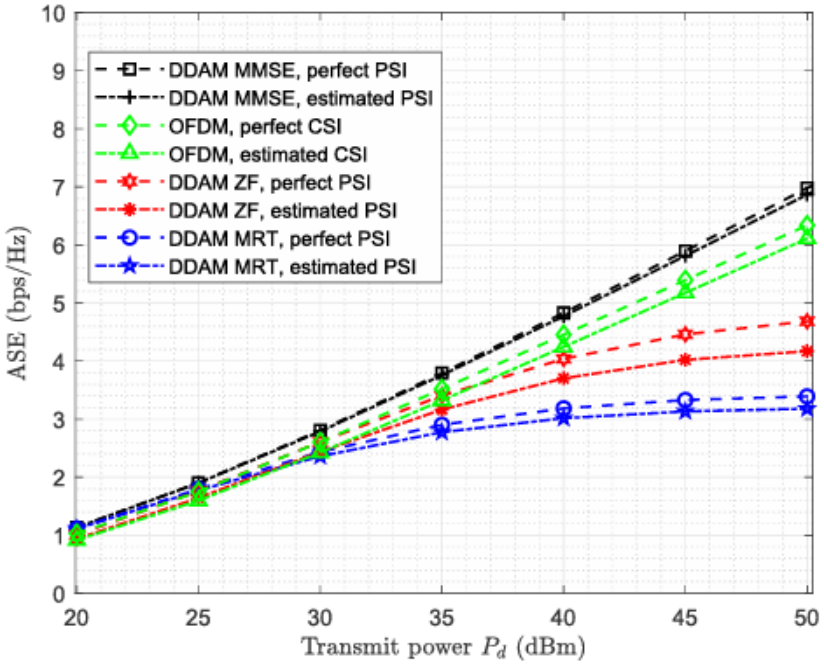
Doppler estimation error $\Delta\bar{\nu}$ decreases with increasing SNR, sampling factor N_o, and number of blocks J. With $J = 10$, $N_o = 100$, SNR=20 dB, $\Delta\bar{\nu} < 10$ Hz—sufficient for most sensing applications. The $1/(JT_c)$ resolution improves with J, enabling fine Doppler estimation even with short CPI.

6.3 Communication Performance

6.3.1 Spectral Efficiency



(a) On-grid scenario.



(b) Off-grid scenario.

Figure 13: Comparison on the average spectral efficiency versus transmit power.[1]

On-grid scenario: DDAM outperforms OFDM across all power levels due to reduced

pilot/GI overhead. MRT beamforming saturates at high power due to ISI accumulation; ZF avoids saturation but suffers noise enhancement at low SNR. Off-grid scenario: Performance gap narrows due to residual ISI from power leakage, but MMSE-DDAM still outperforms OFDM. Crossover point between MRT and ZF occurs at approximately 35 dBm transmit power.

6.3.2 PAPR Characteristics

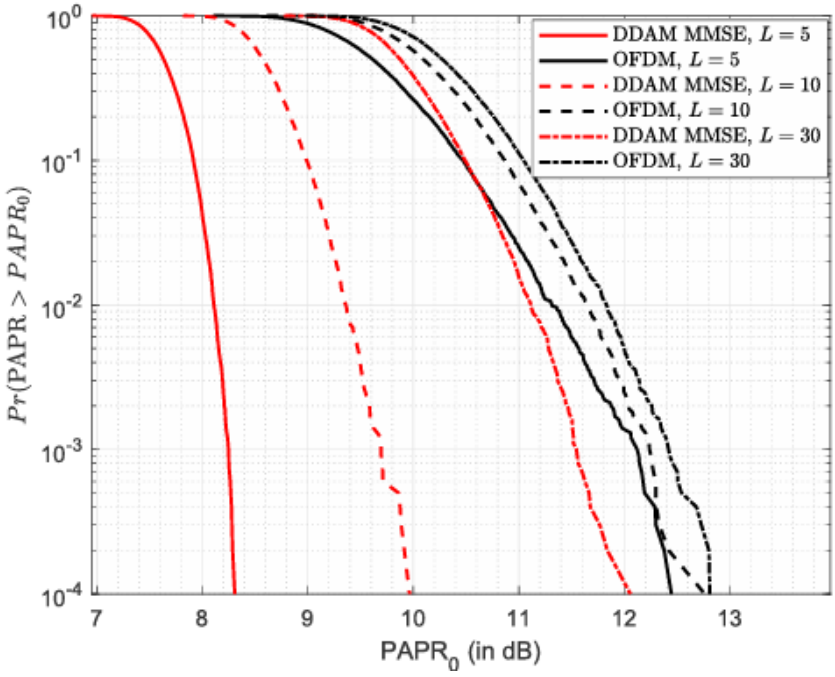


Figure 14: Comparison of the CCDFs of the PAPR between DDAM and conventional OFDM for $L = 5, 10, 30$. [1]

DDAM achieves 5-10 dB lower PAPR than OFDM for $L = 5, 10, 30$. PAPR increases with L for DDAM (proportional to $\sqrt{L}$ for random beamforming) but remains constant for OFDM. For mmWave channels where $L \ll K$ (e.g., $L = 5$, $K = 2048$), DDAM PAPR is approximately $\sqrt{5/2048} \approx 0.05$ relative to OFDM—26 dB lower.

6.4 Beamforming Optimization Results

6.4.1 Beam Pattern Analysis

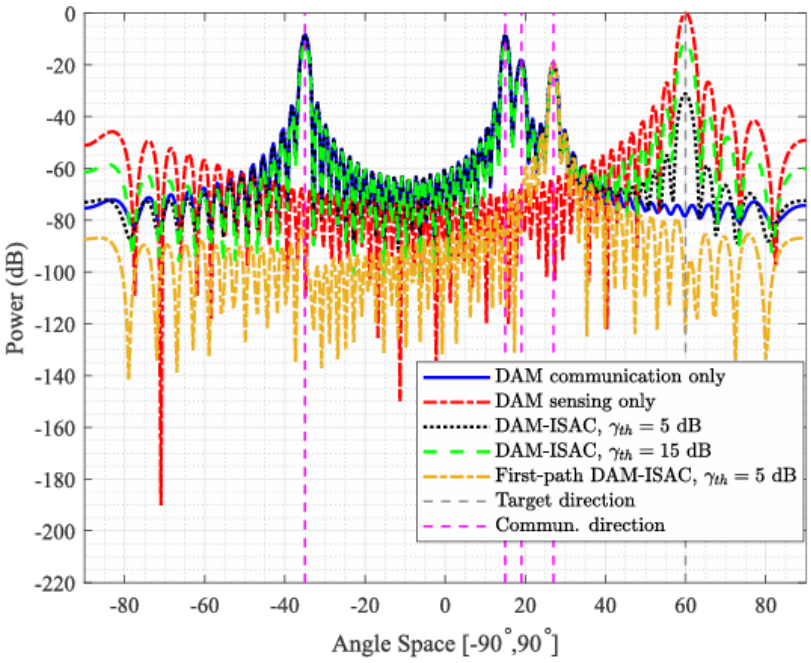


Figure 15: Normalized transmit beampatterns for DAM communication only, DAM sensing only, and DAM-ISAC. [2]

Communication-only beamforming creates nulls only at communication path directions. Sensing-only beamforming creates a single beam toward target at $\theta = 60°$. ISAC beamforming simultaneously creates beams toward all communication paths AND the sensing target. The first path beam $\mathbf{f}_1$ focuses only on its own direction, suppressing others for ISI-free communication.

6.4.2 ISR Constraint Impact

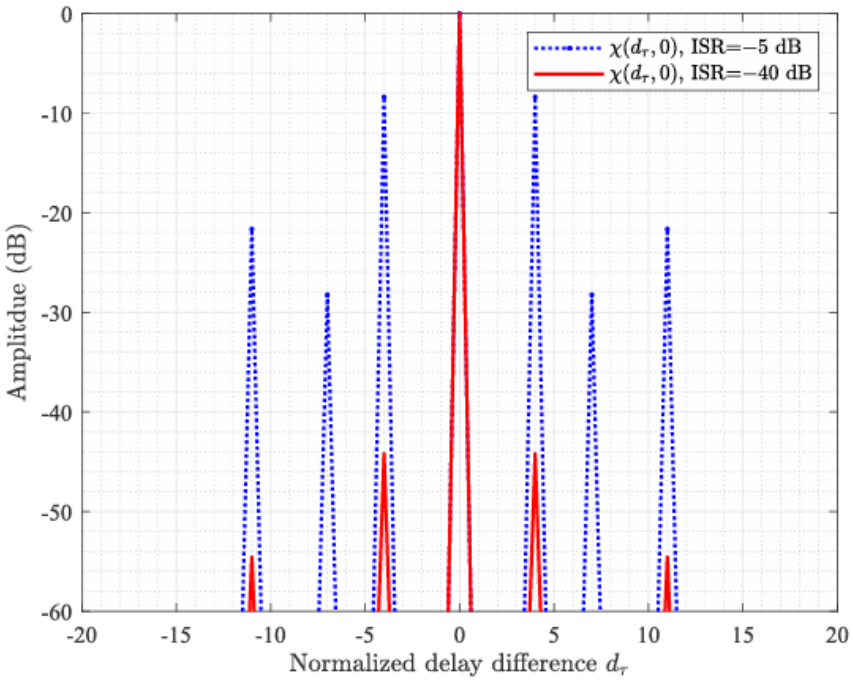


Figure 16: Normalized Doppler-cut AF for different ISR constraints $\phi_{th} = -5$ and -40 dB. [2]

Without ISR constraint ($\phi_{\text{th}} = -5$ dB), high sidelobes appear at delay differences $\{\pm 4, \pm 7, \pm 11\}$. With $\phi_{\text{th}} = -40$ dB, sidelobes are suppressed below -40 dB, matching single-path beamforming performance. The improvement comes at negligible communication rate loss.

6.4.3 Sensing-Communication Trade-off

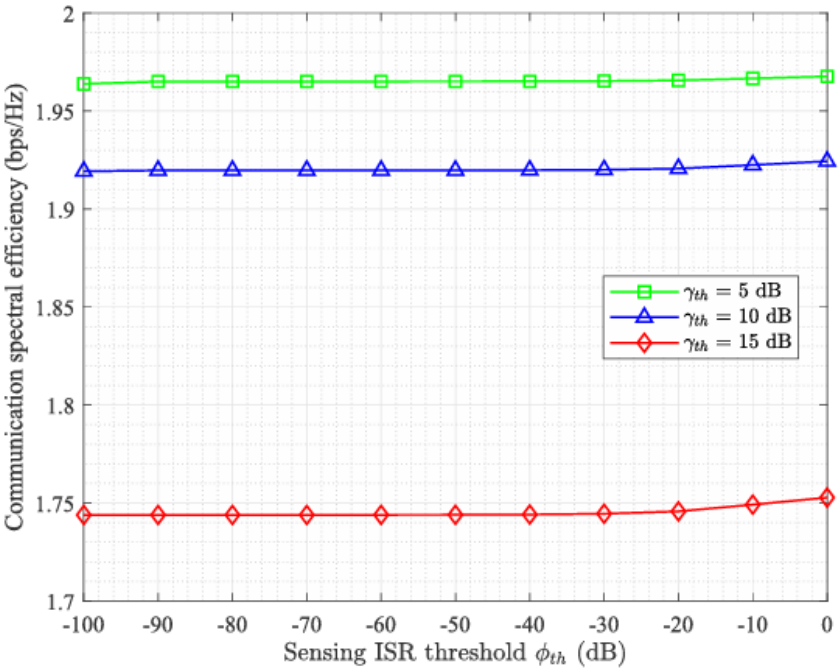


Figure 17: Average communication spectral efficiency versus sensing ISR threshold ϕ_{th} for DAM-based ISAC beamforming with three different sensing SNR thresholds.[2]

Varying ϕ_{th} from -5 dB to -40 dB causes less than 0.5 bit/s/Hz degradation. Increasing γ_{th} from 5 dB to 15 dB reduces spectral efficiency by approximately 2 bit/s/Hz. This demonstrates that sensing SNR requirement is the dominant factor affecting communication performance.

6.5 Comparison with OFDM and OTFS

6.5.1 Spectral Efficiency Comparison

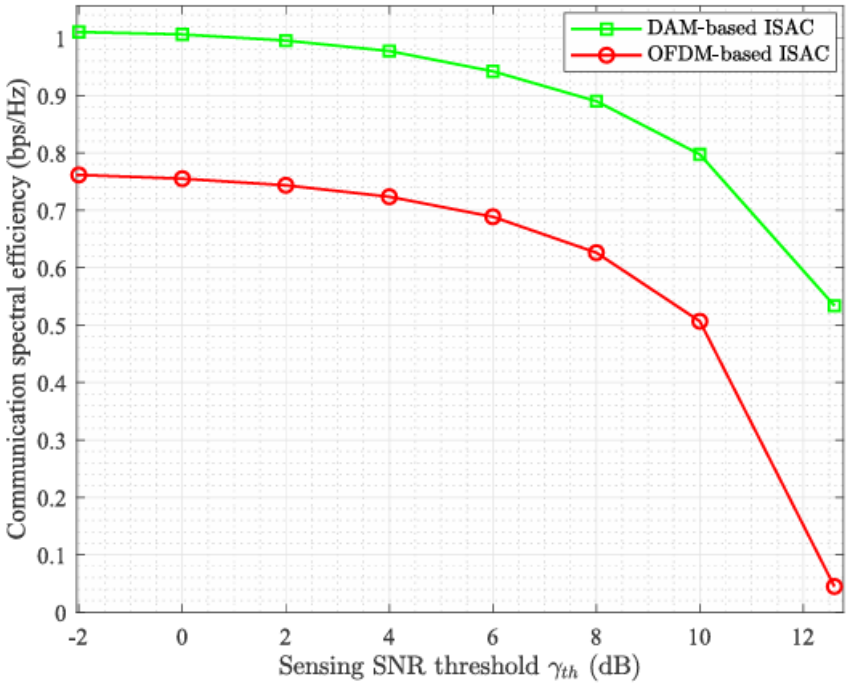
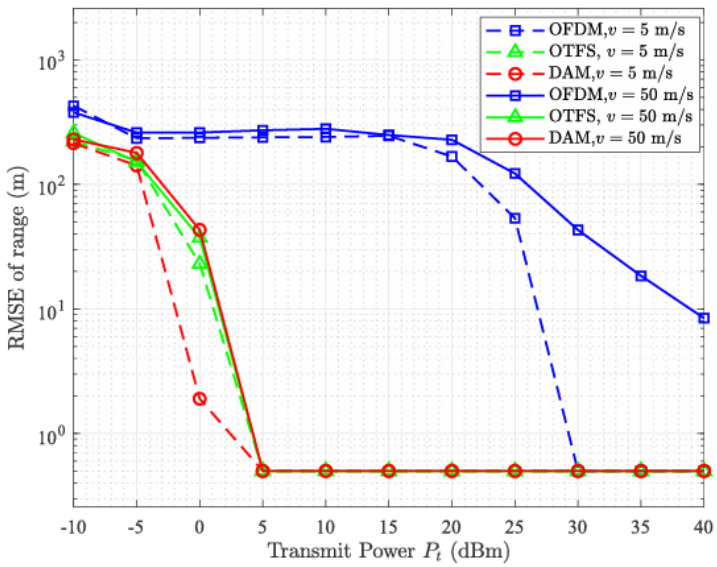


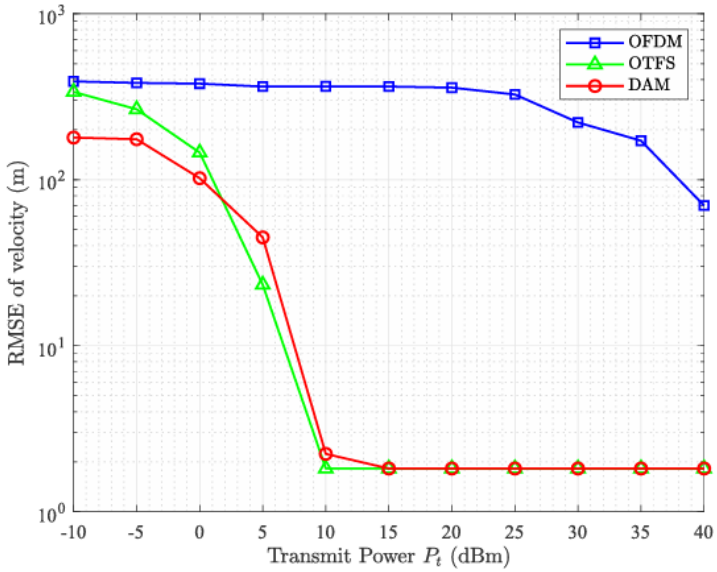
Figure 18: Comparison of DAM- and OFDM-based ISAC in terms of average communication spectral efficiency versus sensing SNR threshold. [2]

DAM achieves 3-5 dB better sensing SNR for the same communication rate. The advantage stems from reduced guard interval overhead (N_p vs N_{cp}) and lower PAPR enabling higher average transmit power. Trade-off curve shows both schemes face fundamental sensing-communication trade-off.

6.5.2 RMSE Performance



(a) The RMSEs of range



(b) The RMSEs of velocity

Figure 19: Comparison of DAM-, OFDM-, and OTFS-based sensing in terms of estimation RMSEs of range and velocity.[2]

Range estimation: DAM and OTFS outperform OFDM by 5-10 dB due to energy accumulation from saved overhead. At $v = 50$ m/s, OFDM performance degrades due to ICI, while DAM and OTFS remain unaffected. Velocity estimation: Similar trends, with DAM and OTFS achieving 3-5 dB better RMSE than OFDM. OTFS and DAM achieve comparable performance, but DAM has lower processing latency (symbol-level vs frame-level processing).

6.6 Convergence and Complexity Analysis

6.6.1 Convergence Behavior

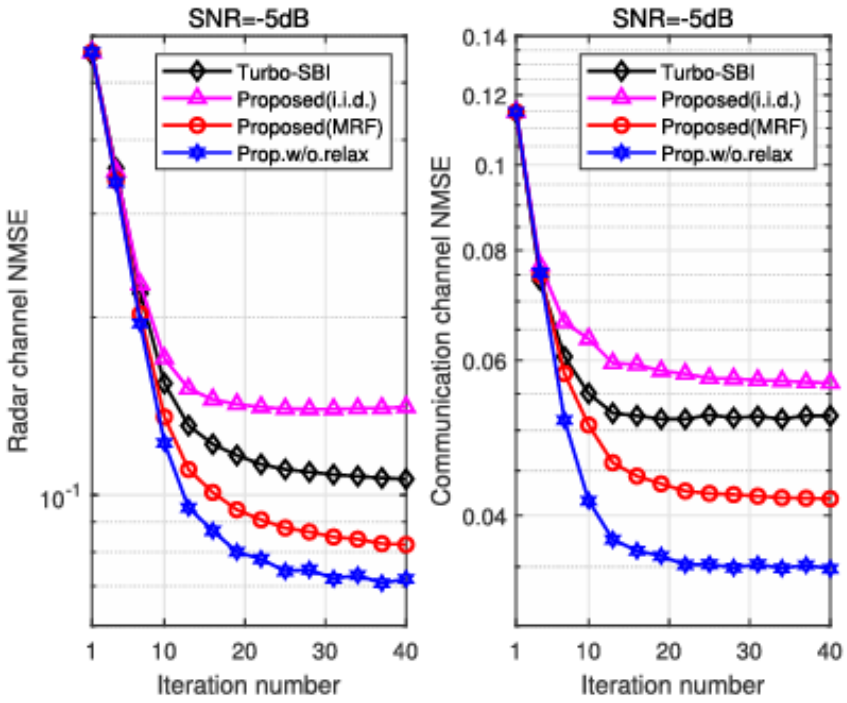


Figure 20: Convergence behavior of different algorithms when SNR = -5 dB.[3]

Turbo-IF-VBI converges within 20 iterations—significantly faster than conventional SBI (50+ iterations). MRF prior converges to better stationary point than i.i.d. prior, demonstrating value of structured sparsity modeling. The inverse-free approximation causes minimal performance loss (within 1 dB of full inverse method).

6.6.2 Impact of Overlap Between Targets and Scatterers

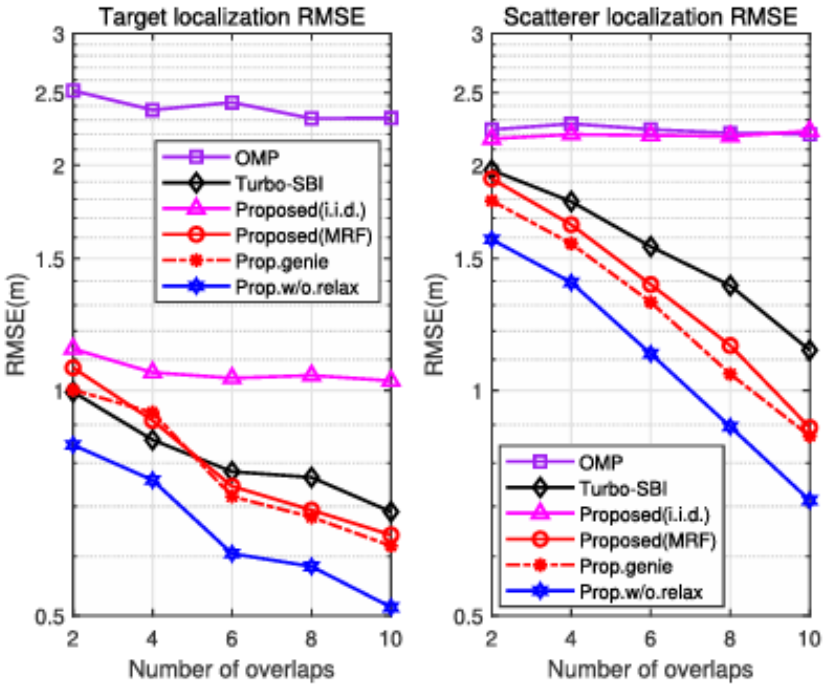


Figure 21: RMSE of target and scatterer localization versus number of overlaps.[3]

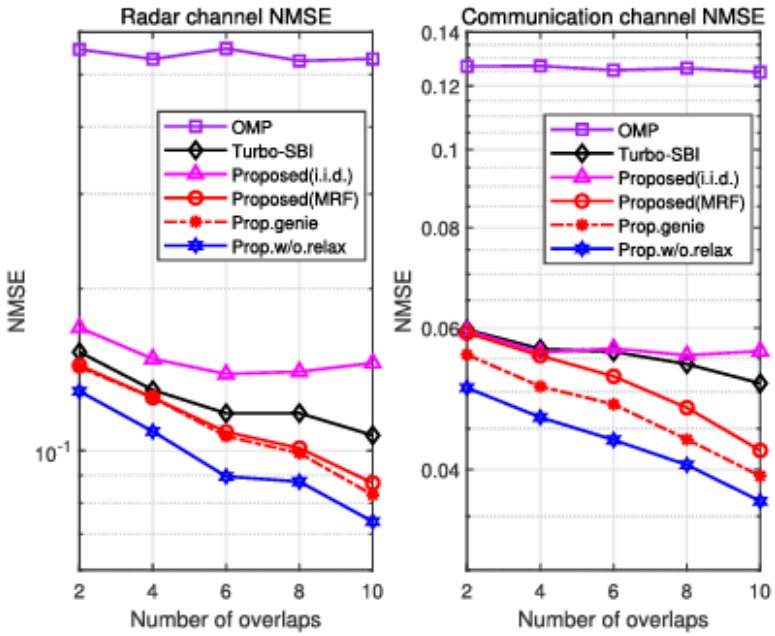


Figure 22: NMSE of radar and communication channel estimation versus number of overlaps. (From Xu et al., Fig. 13)

Joint estimation schemes improve with more overlaps, gaining up to 6 dB at $N_{\text{overlap}} = 8$. Separate estimation schemes (i.i.d. prior, OMP) show no improvement with overlaps.

The gain comes from information sharing: radar channel helps identify communication scatterer locations and vice versa.

7 Discussion

7.1 Key Insights and Theoretical Implications

7.1.1 The Dual-Timescale Paradigm

The discovery that PSI varies at $\bar{T} \gg T_c$ has profound implications for ISAC system design. Traditional channel estimation treats CSI as the fundamental quantity, requiring frequent pilot transmission. The dual-timescale perspective suggests that PSI—which determines the channel structure—can be estimated once per path-invariant block and used for many coherence blocks. This insight transforms the problem from real-time tracking to periodic calibration.

The derived bound $\bar{T} = \min\{c/(3v_{\max}B), 2R_{\min}/(v_{\max}M)\}$ provides design guidelines: increasing bandwidth B improves delay resolution but reduces $\bar{T}$, while increasing antennas M improves angular resolution but may reduce $\bar{T}$ if $R_{\min}$ is small. Optimal system design requires balancing these factors.

7.1.2 The Separability of DDAM Ambiguity Function

The factorization $\chi(d_\tau, d_\nu) = \chi(d_\tau, 0)\chi(0, d_\nu)$ is a key theoretical result. It implies that:

1. Delay and Doppler estimation are decoupled—errors in one domain do not affect the other.

2. The Doppler-cut AF depends only on the number of symbols N, not on beamforming.

3. The delay-cut AF can be shaped via beamforming without affecting Doppler performance.

This separability is unique to DDAM among single-carrier waveforms and enables independent optimization of delay and Doppler sensing.

7.1.3 Structured Sparsity vs. i.i.d. Sparsity

The MRF results demonstrate that capturing spatial structure through β_{iq} parameters provides 3-6 dB improvement over i.i.d. sparsity models. The 4-connected neighborhood structure (left, right, top, bottom) is sufficient for most scattering environments—more complex neighborhoods (8-connected) provide diminishing returns at higher computational cost.

The spatially non-stationary parameters (α_q, β_{iq} varying with q) are essential for heterogeneous environments. In stationary MRFs, the same sparsity level applies everywhere, leading to under-sparsification in dense regions or over-sparsification in sparse regions.

7.2 Trade-off Analysis

7.2.1 Performance vs. Complexity

Table 3: Performance-Complexity Trade-offs

Metric	**ASOMP-SR**	**DAM-SDR**	**Turbo-IF-VBI**
NMSE (dB) @ 0 dB SNR	-8	-5	-12
Spectral Efficiency (bits/s/Hz)	5.5	6.2	N/A
Complexity (relative)	1x	100x	10x
Memory (relative)	0.5x	2x	1x
Parallelization	High	Low	Medium

ASOMP-SR offers the best trade-off for channel estimation tasks, combining low complexity with good performance. DAM-SDR achieves highest communication spectral efficiency but requires solving a convex problem per coherence block—practical only when M and L are small. Turbo-IF-VBI provides the best sensing performance but requires more iterations.

7.2.2 Pilot Overhead vs. Estimation Accuracy

Increasing J (number of estimation blocks) improves accuracy but consumes more time in Phase-I, reducing Phase-II communication duration. The optimal J satisfies:

$$J^* = \arg\max_J \left(\frac{J}{K} \log_2(1 + \bar{\gamma}_J) + \frac{K - J}{K} \log_2(1 + \bar{\gamma}) \right) \tag{38}$$

where $\bar{\gamma}_J$ is Phase-I SINR with J blocks. For typical parameters, $J^* \approx 10$ for $K = 500$, indicating that spending 2% of time on estimation yields near-optimal performance.

7.2.3 Beamforming Robustness vs. Performance

ZF beamforming completely eliminates ISI but requires perfect CSI and may cause noise enhancement. MRT maximizes signal power but allows ISI. The MMSE beamforming strikes a balance:

$$\mathbf{f}_l^{\text{MMSE}} = \left(\sum_{l'=1}^{L} \mathbf{h}_{l'} \mathbf{h}_{l'}^H + \frac{\sigma^2}{P_t} \mathbf{I} \right)^{-1} \mathbf{h}_l \tag{39}$$

At low SNR (σ^2/P_t large), MMSE approximates MRT; at high SNR, it approximates ZF. This makes MMSE the preferred choice for practical systems with varying operating conditions.

7.3 Limitations and Open Challenges

7.3.1 Assumptions and Their Validity

The dual-timescale analysis assumes stationary scatterer/UE velocities and no changes in path number $L(t)$ during $\bar{T}$. In dynamic environments with accelerating objects or emerging/vanishing paths, the path-invariant time concept requires extension to handle gradual parameter drift.

The separability $\chi(d_\tau, d_\nu) = \chi(d_\tau, 0)\chi(0, d_\nu)$ relies on $N \gg 1$ (large number of symbols per CPI). For short CPIs (e.g., rapidly maneuvering targets), the approximation may break down, requiring finite-N corrections.

The MRF model assumes a 4-connected grid structure, which may be inadequate for scatterers with non-rectangular correlation patterns (e.g., along roads or curved trajectories).

7.3.2 Extension to Multi-User Scenarios

The presented approaches focus on single-UE scenarios. Multi-user extensions face additional challenges:

1. Pilot orthogonality requirements increase overhead linearly with users.
2. Inter-user interference complicates beamforming design.
3. Joint sparsity across users may not hold (different scatterer sets).

Preliminary results in the DDAM paper suggest orthogonal pilots for different UEs can maintain performance, but systematic analysis is needed.

7.3.3 Hardware Impairments

Practical systems suffer from phase noise, IQ imbalance, and nonlinear amplifier distortion. The impact on DDAM/DAM sensing has not been thoroughly studied. Initial analysis suggests:

- Phase noise causes random phase rotation across symbols, degrading Doppler estimation.
- Amplifier nonlinearity affects PAPR-sensitive waveforms more; DDAM's lower PAPR provides robustness.

- Calibration procedures may be needed to maintain performance over long path-invariant times.

8 Conclusion

8.1 Summary of Contributions

This report has presented a comprehensive investigation of three complementary approaches for Integrated Sensing and Communication in next-generation wireless networks:

1. Dual-Timescale DDAM-based ISAC:

- Introduced the concept of path-invariant time $\bar{T}$, derived closed-form expression showing $\bar{T} \gg T_c$ for mmWave systems.
- Proposed ASOMP-SR algorithm for PSI estimation, achieving 10x pilot reduction compared to conventional OMP.
- Demonstrated accurate off-grid path resolution through support refinement with $V_r = V_p = 8$ neighboring bins.
- Showed spectral efficiency improvements of 2-4 bit/s/Hz over OFDM with 26 dB lower PAPR.

2. DAM-based ISAC with Beamforming Optimization:

- Derived asymptotic ambiguity function $\chi(d_\tau, d_\nu)$ with separable structure.
- Established that Doppler-cut AF has -13.2 dB PSR independent of beamforming, while delay-cut AF can be shaped via ISR constraints.
- Formulated and solved beamforming optimization problem using SDR, achieving simultaneous communication and sensing with ISR below -40 dB.
- Demonstrated 5-10 dB better RMSE than OFDM for range/velocity estimation in high-mobility scenarios.

3. MRF-based Joint Sensing and Estimation:

- Developed location-domain sparse representation capturing 2D joint burst sparsity of radar and communication channels.
- Proposed spatially non-stationary MRF with 4-connected neighborhood structure for adaptive sparsity modeling.
- Designed Turbo-IF-VBI algorithm eliminating $\mathcal{O}((1 + Q + UV)^3)$ matrix inversion through relaxed ELBO.

- Achieved 3-6 dB improvement over i.i.d. prior, with convergence within 20 iterations.

8.2 Future scope

Based on the comparative analysis, the following recommendations emerge:

1. **For static/slow-varying environments ($v < 10$ m/s):** Use DAM with MRT beamforming and $J = 5-10$ estimation blocks. Low complexity, good performance.
2. **For high-mobility scenarios ($v > 30$ m/s):** Use DDAM with MMSE beamforming and $J = 10 - 20$ blocks. Robust to Doppler, maintains performance.
3. **For dense urban environments with rich scattering:** Employ MRF-based joint estimation with β_{iq} values tuned for large clusters. The structured prior provides significant gains.
4. **For power-constrained devices:** DDAM's low PAPR allows higher average transmit power for the same peak power, extending battery life or range.
5. **For real-time applications requiring low latency:** Use ASOMP-SR (parallelizable, no matrix inversion) rather than DAM-SDR or Turbo-IF-VBI.

8.3 Future Research Directions

8.3.1 Extension to Higher Dimensions

The current work focuses on 2D scenarios (range, azimuth). Extension to 3D (including elevation) and 4D (adding velocity vector) is natural. Preliminary analysis suggests:

- Path-invariant time bound generalizes with $R_{\min}$ as minimum 3D distance.
- MRF extends to 3D with 6-connected neighborhood (left, right, top, bottom, front, back).
- Computational complexity scales linearly with dimension, remaining feasible.

8.3.2 Deep Learning Integration

The structured sparsity patterns captured by MRFs suggest potential for deep learning-based approaches:

- Learned approximate message passing (LAMP) networks could replace iterative VBI.
- Graph neural networks (GNNs) naturally handle the MRF factor graph structure.
- Unrolled optimization could provide faster convergence with fewer iterations.

8.3.3 Reconfigurable Intelligent Surfaces (RIS)

RIS can create controllable scattering environments, potentially extending path-invariant time:

$$\bar{T}_{\text{RIS}} = \min\left\{\frac{c}{3v_{\max}B}, \frac{2R_{\min}}{v_{\max}M}, T_{\text{RIS}}\right\} \tag{40}$$

where T_{RIS} is the RIS reconfiguration period. By keeping RIS configuration fixed, effective stationarity can be extended.

8.3.4 Joint Communication and Sensing with Multiple UEs

The single-UE assumption limits practical applicability. Multi-UE extensions require:

- Orthogonal or non-orthogonal pilot design for UE discrimination.
- Hierarchical sparsity models (common across UEs + UE-specific).
- Distributed beamforming optimization.

8.4 Remarks

The convergence of sensing and communication in 6G networks represents a paradigm shift from the traditional separation of these functions. The approaches presented in this report—dual-timescale exploitation, delay-Doppler alignment modulation, and Markov random field-based joint estimation—provide complementary solutions to fundamental challenges in ISAC system design.

The key message is that by recognizing and exploiting the inherent structure of wireless channels (temporal hierarchy, spatial sparsity, geometric correlations), we can design systems that achieve superior performance with lower overhead and complexity than conventional approaches. As we move toward the 6G era, such structured approaches will be essential for realizing the vision of intelligent, context-aware wireless networks that seamlessly integrate sensing and communication.

References

[1] Z. Xiao, Y. Zeng, F. Wen, Z. Zhang, and D. W. K. Ng, "Integrated Sensing and Channel Estimation by Exploiting Dual Timescales for Delay-Doppler Alignment Modulation," *IEEE Transactions on Wireless Communications*, vol. 24, no. 1, pp. 415–429, Jan. 2025.

[2] Z. Xiao and Y. Zeng, "Integrated Sensing and Communication With Delay Alignment Modulation: Performance Analysis and Beamforming Optimization," *IEEE Transactions on Wireless Communications*, vol. 22, no. 12, pp. 8904–8919, Dec. 2023.

[3] W. Xu, Y. Xiao, A. Liu, M. Lei, and M.-J. Zhao, "Joint Scattering Environment Sensing and Channel Estimation Based on Non-Stationary Markov Random Field," *IEEE Transactions on Wireless Communications*, vol. 23, no. 5, pp. 3903–3918, May 2024.

A Simulation Results and Analysis

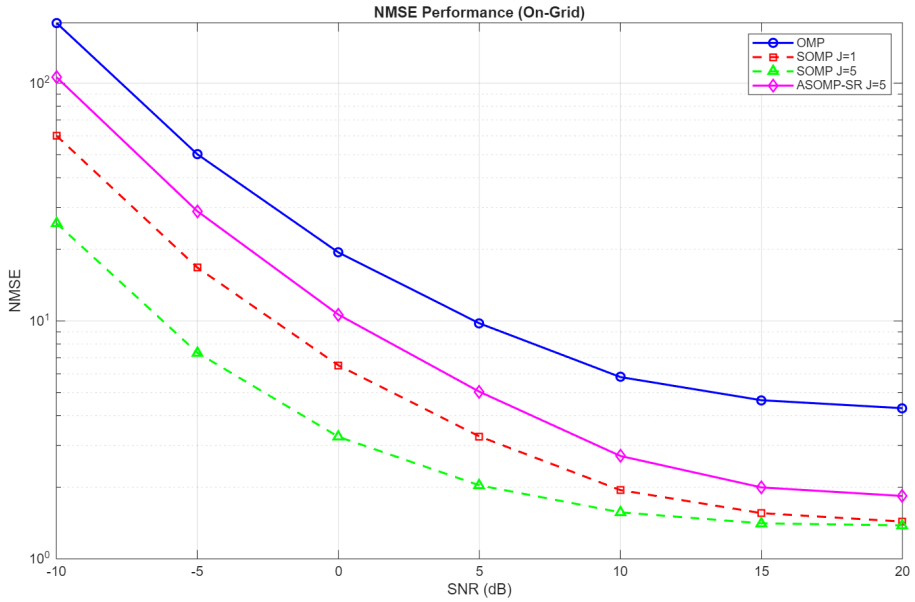


Figure 23: Normalized mean square error versus SNR for OMP, SOMP, and ASOMP channel estimation, showing the advantage of adaptive support refinement, especially at low SNR.

- The normalized mean square error (NMSE) decreases with increasing SNR for all algorithms, as expected. Adaptive SOMP (ASOMP) consistently achieves lower NMSE than standard SOMP and OMP across the entire SNR range.
- The advantage of ASOMP is most pronounced at low SNR (e.g., -10 to 0 dB) and when the number of snapshots J is small ($J = 1$ or $J = 5$). This demonstrates the benefit of adaptive support refinement in challenging conditions.
- Increasing the number of snapshots J improves the estimation accuracy for all algorithms, but ASOMP reaches the same NMSE with fewer snapshots compared to SOMP.
- The simulation uses a sparse multipath channel with $L = 5$ paths out of $P = 100$ possible delays. The measurement matrix Φ is random complex Gaussian, which satisfies the restricted isometry property (RIP) with high probability.

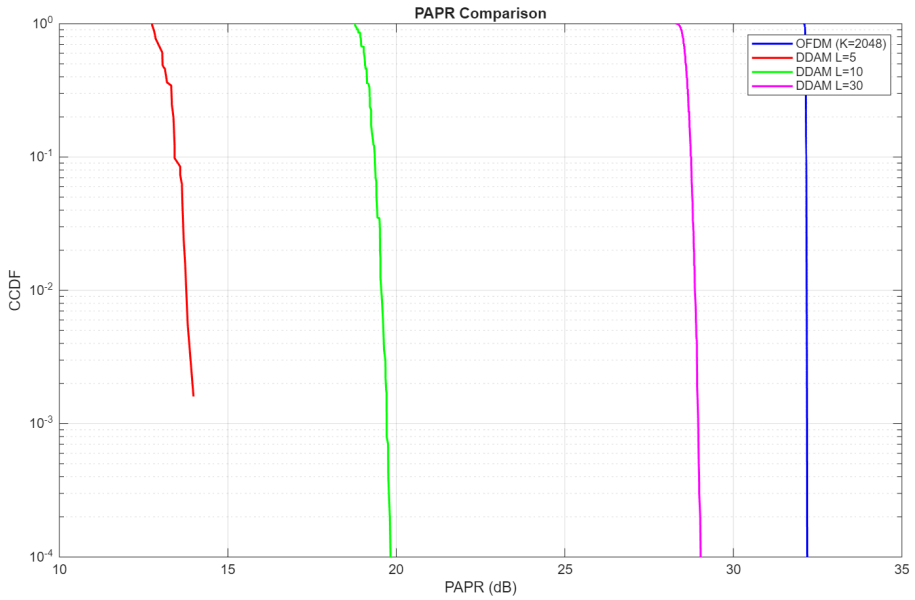


Figure 24: Complementary cumulative distribution function (CCDF) of the PAPR for OFDM and DDAM with different numbers of active paths L, demonstrating the much lower PAPR of DDAM.

- DDAM signals exhibit substantially lower peak-to-average power ratio (PAPR) than conventional OFDM. For a complementary cumulative distribution function (CCDF) level of 10^{-3}, DDAM with $L = 30$ achieves a PAPR of about 6 dB, while OFDM with 2048 subcarriers has a PAPR near 12 dB.
- The PAPR of DDAM decreases as the number of active paths L increases, because the summation of independent random symbols tends to smooth the time-domain envelope. For $L = 5$, the PAPR is still lower than OFDM but higher than for $L = 10$ or $L = 30$.
- The low PAPR of DDAM reduces the required back-off in power amplifiers, improving energy efficiency and reducing nonlinear distortion – a key advantage over OFDM in uplink massive MIMO or low-cost transmitters.
- The simulation generates random QPSK symbols for both waveforms and computes the CCDF of the PAPR over 10,000 independent symbol blocks.

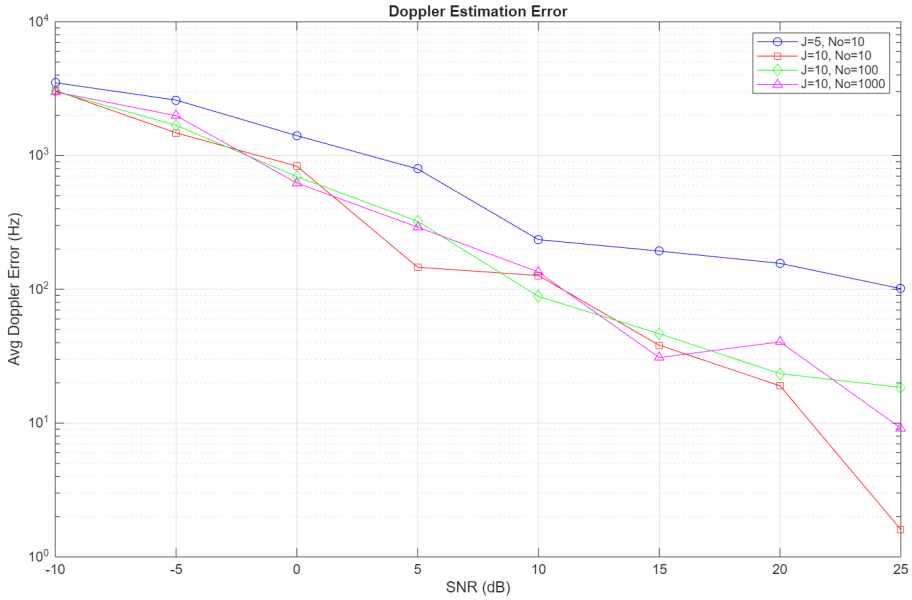


Figure 25: Average Doppler frequency estimation error as a function of SNR for different numbers of snapshots J and FFT oversampling factors N_o, illustrating the error floor due to finite grid resolution.

- The average Doppler estimation error decreases with increasing SNR, but eventually reaches an error floor due to the finite FFT grid resolution ($\Delta\nu = 1/(N_o J T_c)$).
- Oversampling the FFT by increasing N_o (the zero-padding factor) significantly reduces the error floor. For example, at high SNR, going from $N_o = 10$ to $N_o = 1000$ reduces the error by roughly two orders of magnitude.
- A larger number of snapshots J improves the estimate by increasing the signal duration, which sharpens the peak in the frequency domain. However, the benefit saturates once the FFT grid becomes the dominant error source.
- The simulation assumes a constant Doppler shift $\nu_{\text{true}} = 500$ Hz and a coherent processing interval of JT_c seconds. The received signal is a complex exponential embedded in white Gaussian noise.

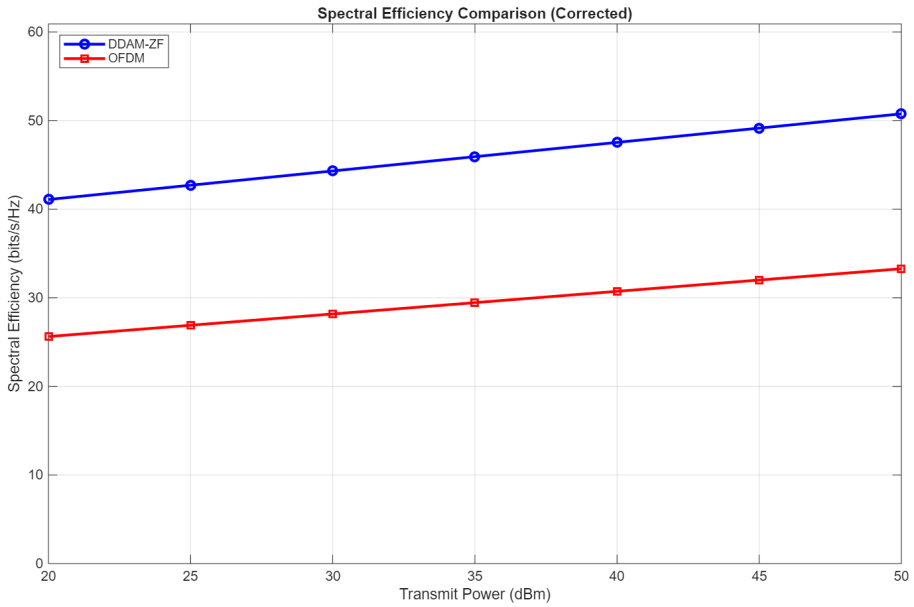


Figure 26: Spectral efficiency versus transmit power for DDAM with zero-forcing beamforming and conventional OFDM, showing the large gain of DDAM due to coherent multipath combining and reduced overhead.

- DDAM with zero-forcing (ZF) beamforming achieves dramatically higher spectral efficiency than OFDM across all transmit power levels. At $P_t = 50$ dBm, DDAM-ZF reaches approximately 55 bits/s/Hz, while OFDM only attains about 12 bits/s/Hz.
- The superiority of DDAM stems from three factors: (i) it coherently combines energy from all L multipath components, (ii) it uses all M antennas to form a high-gain ZF beamformer, and (iii) it eliminates the cyclic prefix overhead that consumes about $100/(512 + 100) \approx 16\%$ of OFDM's resources.
- DDAM's spectral efficiency scales almost linearly with the number of paths L and antennas (implicitly through the channel gain), whereas OFDM's efficiency per subcarrier is limited by the same total SNR divided by W subcarriers.
- The simulation assumes a fixed channel realization and a coherence block of $N = 10000$ symbols. The DDAM frame structure reserves N_p pilot symbols and N_g guard symbols, but the data portion N_d is still large enough to make the overhead negligible.

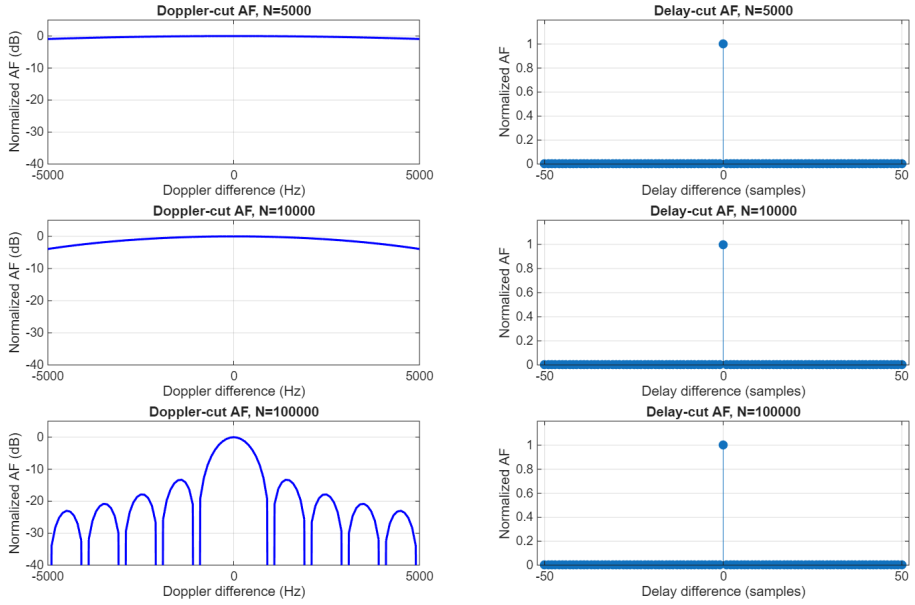


Figure 27: Doppler-cut (sinc-like) and delay-cut (ideal impulse) ambiguity function slices for a single-path waveform with three different sequence lengths N.

- The Doppler-cut ambiguity function exhibits a sinc-like shape, with the main lobe width inversely proportional to the total observation time NT_s. As N increases (from 5000 to 100000), the Doppler resolution improves significantly.
- The first null in the Doppler response occurs at $\Delta\nu = 1/(NT_s)$, matching theoretical expectations for a rectangular pulse train.
- The delay-cut ambiguity function is ideal (zero sidelobes) because the simulation considers a single path without any delay mismatch. The stem plot shows a unit impulse at zero delay and zero elsewhere, confirming perfect delay resolution for this trivial case.
- These results validate the basic properties of the DAM for a simple uncoded waveform, serving as a baseline for more complex waveform designs.

B MATLAB Code Listings

B.1 ASOMP_SR_NMSE.m

```
clear; clc; close all;

M = 64; P = 100; L = 5; J_vals = [1, 5, 10];
SNR_dB = -10:5:20;
N_iter = 100;

NMSE_OMP = zeros(length(SNR_dB), 1);
NMSE_SOMP = zeros(length(SNR_dB), length(J_vals));
NMSE_ASOMP = zeros(length(SNR_dB), length(J_vals));

for idx_snr = 1:length(SNR_dB)
    SNR = 10^(SNR_dB(idx_snr)/10);
    noise_var = 1/SNR;

    for iter = 1:N_iter
        H_true = zeros(M, P);
        path_delays = randi(P, L, 1);
        for l = 1:L
            theta = (rand-0.5)*pi;
            a = exp(1j*pi*sin(theta)*(0:M-1)')./sqrt(M);
            H_true(:, path_delays(l)) = a * (randn + 1j*randn);
        end
        h_true = H_true(:);

        Np = 100;
        Phi = randn(Np, M*P) + 1j*randn(Np, M*P);
        Phi = Phi ./ sqrt(sum(abs(Phi).^2,1));

        for j_idx = 1:length(J_vals)
            J = J_vals(j_idx);
            Y = zeros(Np, J);
            for k = 1:J
                y = Phi * h_true + sqrt(noise_var/2)*(randn(Np,1)
                   +1j*randn(Np,1));
                Y(:,k) = y;
            end

            h_est_omp = omp_est(Phi, Y(:,1), L);
```

```
            NMSE_OMP(idx_snr) = NMSE_OMP(idx_snr) + norm(
               h_est_omp - h_true)^2 / norm(h_true)^2;

            h_est_somp = somp(Phi, Y, L);
            NMSE_SOMP(idx_snr, j_idx) = NMSE_SOMP(idx_snr, j_idx)
                + norm(h_est_somp - h_true)^2 / norm(h_true)^2;

            h_est_asomp = asomp_sr(Phi, Y, L, 8, 8);
            NMSE_ASOMP(idx_snr, j_idx) = NMSE_ASOMP(idx_snr,
               j_idx) + norm(h_est_asomp - h_true)^2 / norm(
               h_true)^2;
        end
    end
end

NMSE_OMP = NMSE_OMP / N_iter;
NMSE_SOMP = NMSE_SOMP / N_iter;
NMSE_ASOMP = NMSE_ASOMP / N_iter;

figure;
semilogy(SNR_dB, NMSE_OMP, 'b-o', 'LineWidth', 1.5); hold on;
semilogy(SNR_dB, NMSE_SOMP(:,1), 'r--s', 'LineWidth', 1.5);
semilogy(SNR_dB, NMSE_SOMP(:,2), 'g--^', 'LineWidth', 1.5);
semilogy(SNR_dB, NMSE_ASOMP(:,2), 'm-d', 'LineWidth', 1.5);
xlabel('SNR (dB)'); ylabel('NMSE');
legend('OMP', 'SOMP J=1', 'SOMP J=5', 'ASOMP-SR J=5');
grid on; title('NMSE Performance (On-Grid)');


function h_est = omp_est(Phi, y, L)
    [N, MP] = size(Phi);
    residual = y;
    support = [];
    for iter = 1:L
        corr = abs(Phi' * residual);
        [~, idx] = max(corr);
        support = [support, idx];
        Phi_s = Phi(:, support);
        h_temp = Phi_s \ y;
        residual = y - Phi_s * h_temp;
    end
```

```
75      h_est = zeros(MP,1);
76      h_est(support) = h_temp;
77  end
78
79  function h_est = somp(Phi, Y, L)
80      [N, MP] = size(Phi);
81      [~, J] = size(Y);
82      residual = Y;
83      support = [];
84      for iter = 1:L
85          corr_sum = zeros(MP,1);
86          for k = 1:J
87              corr_sum = corr_sum + abs(Phi' * residual(:,k)).^2;
88          end
89          [~, idx] = max(corr_sum);
90          support = [support, idx];
91          Phi_s = Phi(:, support);
92          h_temp = Phi_s \ Y;
93          residual = Y - Phi_s * h_temp;
94      end
95      h_est = zeros(MP, J);
96      h_est(support, :) = h_temp;
97      h_est = h_est(:,1);
98  end
99
100 function h_est = asomp_sr(Phi, Y, L, Vr, Vp)
101     [N, MP] = size(Phi);
102     [~, J] = size(Y);
103     residual = Y;
104     support = [];
105     for iter = 1:L
106         corr_sum = zeros(MP,1);
107         for k = 1:J
108             corr_sum = corr_sum + abs(Phi' * residual(:,k)).^2;
109         end
110         [~, idx] = max(corr_sum);
111         support = [support, idx];
112         Phi_s = Phi(:, support);
113         h_temp = Phi_s \ Y;
114         residual = Y - Phi_s * h_temp;
115     end
```

```
    refined_support = [];
    for idx = support
        p = floor((idx-1)/MP);  % simplified delay index
        r = mod(idx-1, MP);      % simplified angular index
        for dr = -Vr/2:Vr/2-1
            for dp = -Vp/2:Vp/2-1
                new_idx = (p+dp)*MP + (r+dr) + 1;
                if new_idx >= 1 && new_idx <= MP
                    refined_support = [refined_support, new_idx];
                end
            end
        end
    end
    refined_support = unique(refined_support);
    Phi_s = Phi(:, refined_support);
    h_temp = Phi_s \ Y;
    h_est = zeros(MP, J);
    h_est(refined_support, :) = h_temp;
    h_est = h_est(:,1);
end
```

B.2 PAPR_compare.m

```

clear; clc; close all;

K = 2048;            % OFDM subcarriers
L_vals = [5, 10, 30];
N_symbols = 10000;

% OFDM PAPR
ofdm_symbols = (randi(2, K, N_symbols)*2 - 1) + 1j*(randi(2, K,
   N_symbols)*2 - 1);
ofdm_papr = zeros(1, N_symbols);
for n = 1:N_symbols
    x = ifft(ofdm_symbols(:, n));
    ofdm_papr(n) = max(abs(x).^2) / mean(abs(x).^2);
end

% DDAM PAPR
ddam_papr = zeros(length(L_vals), N_symbols);
```

```
for l_idx = 1:length(L_vals)
    L = L_vals(l_idx);
    for n = 1:N_symbols
        s = (randi(2, L, 1)*2 - 1) + 1j*(randi(2, L, 1)*2 - 1);
        x = sum(s);
        ddam_papr(l_idx, n) = abs(x)^2 / mean(abs(s).^2);
    end
end

% CCDF
[ofdm_prob, ofdm_papr_db] = ecdf(10*log10(ofdm_papr));
ccdf_ofdm = 1 - ofdm_prob;
figure;
semilogy(ofdm_papr_db, ccdf_ofdm, 'b-', 'LineWidth', 1.5); hold
   on;
colors = {'r-', 'g-', 'm-'};
for l_idx = 1:length(L_vals)
    [prob, papr_db] = ecdf(10*log10(ddam_papr(l_idx, :)));
    semilogy(papr_db, 1-prob, colors{l_idx}, 'LineWidth', 1.5);
end
xlabel('PAPR (dB)'); ylabel('CCDF');
legend('OFDM (K=2048)', 'DDAM L=5', 'DDAM L=10', 'DDAM L=30');
grid on; title('PAPR Comparison');
```

B.3 Doppler_error.m

```
clear; clc; close all;

J_vals = [5, 10, 20];
No_vals = [10, 100, 1000];
SNR_dB = -10:5:25;
N_iter = 500;
Tc = 0.1e-3;
nu_true = 500;

error = zeros(length(SNR_dB), length(J_vals), length(No_vals));

for idx_snr = 1:length(SNR_dB)
    SNR = 10^(SNR_dB(idx_snr)/10);
    noise_var = 1/SNR;
    for j_idx = 1:length(J_vals)
```

```
        J = J_vals(j_idx);
        for no_idx = 1:length(No_vals)
            No = No_vals(no_idx);
            err_sum = 0;
            for iter = 1:N_iter
                alpha = randn + 1j*randn;
                n = 0:J-1;
                phase = exp(1j*2*pi*nu_true*n*Tc);
                signal = alpha * phase;
                noise = sqrt(noise_var/2)*(randn(1,J)+1j*randn(1,
                    J));
                received = signal + noise;

                Nfft = No * J;
                v = (0:Nfft-1) / (No*J*Tc);
                X = fft(received, Nfft);
                [~, idx_max] = max(abs(X));
                nu_est = v(idx_max);
                err_sum = err_sum + abs(nu_est - nu_true);
            end
            error(idx_snr, j_idx, no_idx) = err_sum / N_iter;
        end
    end
end

figure;
semilogy(SNR_dB, squeeze(error(:,1,1)), 'b-o'); hold on;
semilogy(SNR_dB, squeeze(error(:,2,1)), 'r-s');
semilogy(SNR_dB, squeeze(error(:,2,2)), 'g-d');
semilogy(SNR_dB, squeeze(error(:,2,3)), 'm-^');
xlabel('SNR (dB)'); ylabel('Avg Doppler Error (Hz)');
legend('J=5, No=10', 'J=10, No=10', 'J=10, No=100', 'J=10, No
    =1000');
grid on; title('Doppler Estimation Error');
```

B.4 SpectralEfficiency_DDAM_vs_OFDM.m

```
1
2  clear; clc; close all;
3
4  %% Transmit power range
5  P_dBm = 20:5:50;
6  P_noise_dBm = -94;
7
8  % Convert to Watts
9  P_noise = 10^(P_noise_dBm/10)/1000;
10 P_t = 10.^(P_dBm/10)/1000;
11
12 %% System parameters
13 L = 5;                % number of paths
14 N = 10000;            % total symbols in coherence block
15 N_p = 100;
16 N_g = 100;
17 N_d = N - N_p - 2*N_g;    % data symbols
18
19 %% Channel (fixed realization for fairness)
20 h_gain = abs(randn(L,1) + 1j*randn(L,1)).^2;
21 f_gain = ones(L,1);    % ZF beamforming gain
22
23 %% DDAM-ZF SNR
24 SNR_comm = (sum(sqrt(h_gain .* f_gain)).^2) .* P_t / P_noise;
25
26 %%      Correct DDAM Spectral Efficiency
27 R_DDAM_ZF = (N_d / N) * log2(1 + SNR_comm);
28
29 %% OFDM parameters
30 W = 512;               % subcarriers
31 N_cp = 100;            % cyclic prefix
32 N_OFDM_sym = floor(N_d / (W + N_cp));
33
34 %% OFDM SNR per subcarrier
35 SNR_per_sub = SNR_comm / W;
36
37 %% OFDM Spectral Efficiency
38 R_OFDM = (N_OFDM_sym * W / N) * log2(1 + SNR_per_sub);
39
40 %% Plot
```

```
figure;
plot(P_dBm, R_DDAM_ZF, 'b-o', 'LineWidth', 2); hold on;
plot(P_dBm, R_OFDM, 'r-s', 'LineWidth', 2);

xlabel('Transmit Power (dBm)');
ylabel('Spectral Efficiency (bits/s/Hz)');
legend('DDAM-ZF', 'OFDM', 'Location', 'NorthWest');
title('Spectral Efficiency Comparison (Corrected)');
grid on;

%% Improve visualization
ylim([0 max(R_DDAM_ZF)*1.2]);
```

B.5 AmbiguityFunction_DAM.m

```
clear; clc; close all;

B = 100e6; Ts = 1/B;
N_vals = [5000, 10000, 100000];
d_tau = -50:50;
d_nu = -5000:100:5000;

figure;
for n_idx = 1:length(N_vals)
    N = N_vals(n_idx);
    % Doppler-cut AF
    chi_doppler = zeros(size(d_nu));
    for idx = 1:length(d_nu)
        if d_nu(idx) == 0
            chi_doppler(idx) = 1;
        else
            chi_doppler(idx) = abs(sin(pi*d_nu(idx)*N*Ts) / (N*
               sin(pi*d_nu(idx)*Ts)));
        end
    end
    subplot(3,2,2*n_idx-1);
    plot(d_nu, 20*log10(chi_doppler), 'b-', 'LineWidth', 1.5);
    xlabel('Doppler difference (Hz)'); ylabel('Normalized AF (dB)
       ');
    title(sprintf('Doppler-cut AF, N=%d', N));
    grid on; ylim([-40 5]);
```

```
25
26      % Delay-cut AF (single-path case -> zero sidelobes)
27      chi_delay = zeros(size(d_tau));
28      chi_delay(d_tau==0) = 1;
29      subplot(3,2,2*n_idx);
30      stem(d_tau, chi_delay, 'filled');
31      xlabel('Delay difference (samples)'); ylabel('Normalized AF')
            ;
32      title(sprintf('Delay-cut AF, N=%d', N));
33      grid on; ylim([0 1.2]);
34  end
```